



Inspiring Excellence

CSE 251

Electronic Devices and Circuits

Lecture 11

Course instructor:

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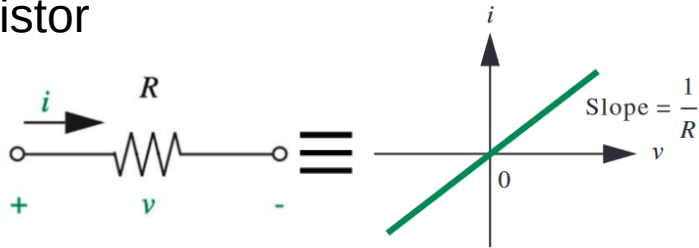
Outline

- Introduction to Electronic Switches
- Basic Inverter
- Introduction to Controlled Sources
- Introduction to MOSFET
- MOSFET as digital switch
- Designing Logic gates with MOSFETs

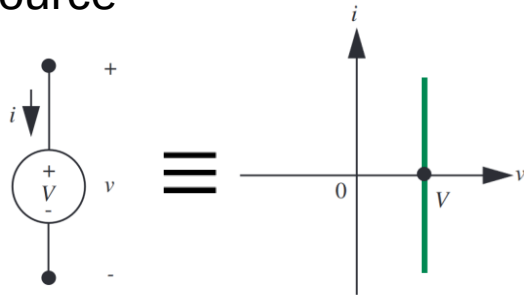
Two terminal Devices

Linear Devices

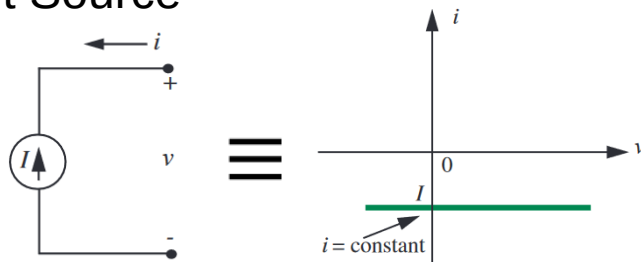
Resistor



Voltage Source



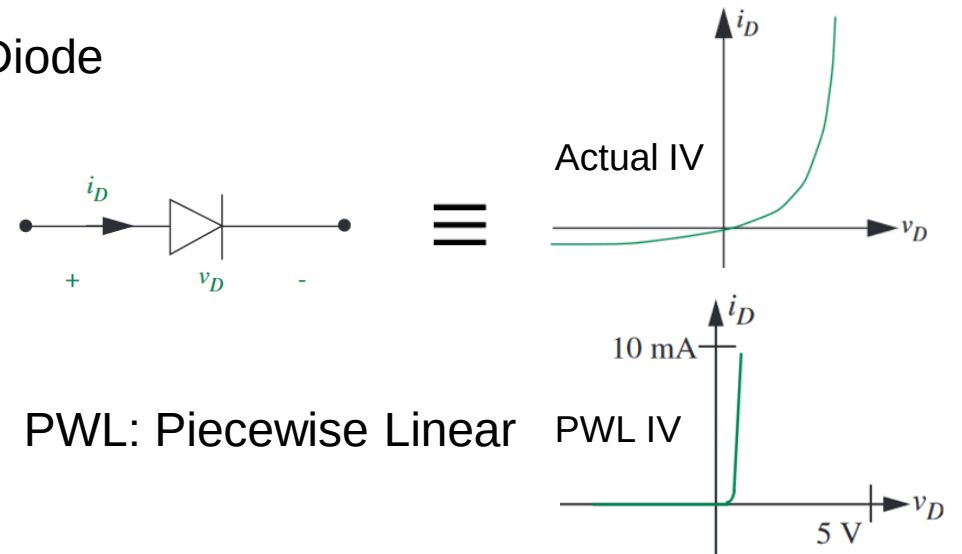
Current Source



IV characteristics of two terminal devices are fixed.

Non-linear Devices

Diode



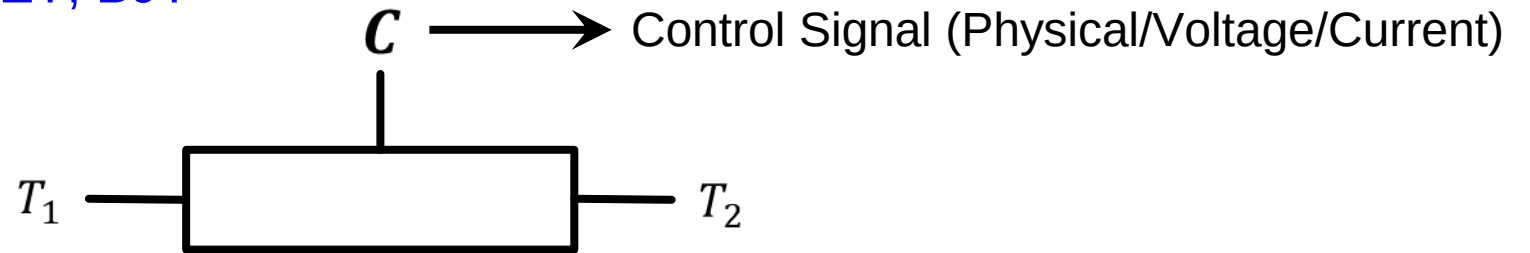
IV characteristics of **three terminal devices** can be changed

Three terminal Devices

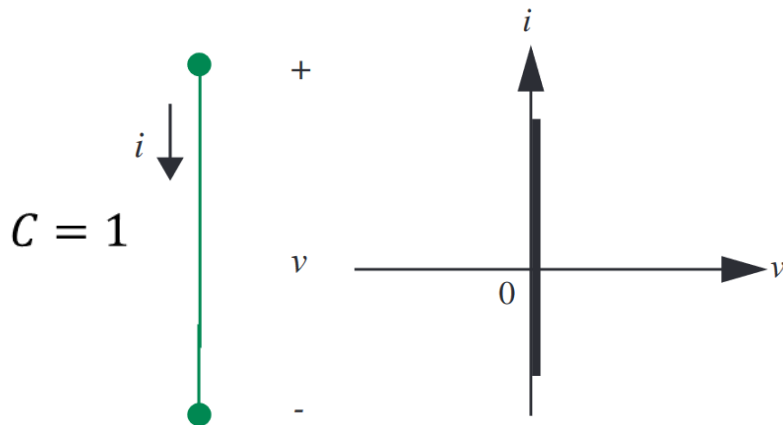
IV of two terminal can be controlled using a third terminal.

Example: Switch, MOSFET, BJT

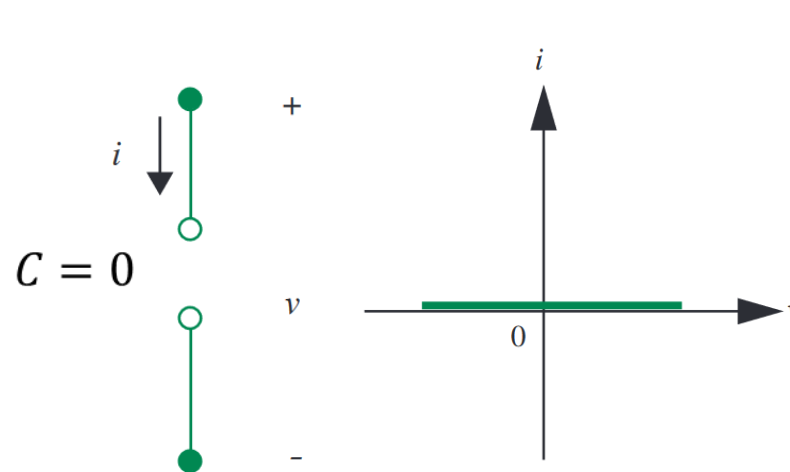
Switch



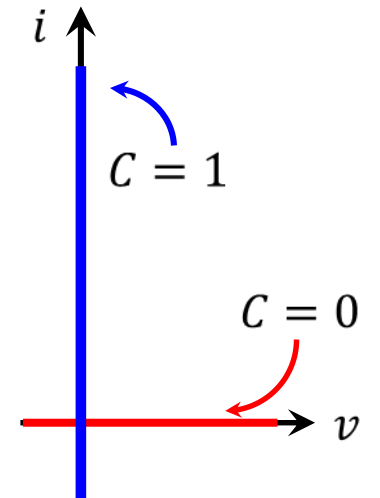
IV characteristics between T_1 and T_2 can be controlled by C



Switch **ON**

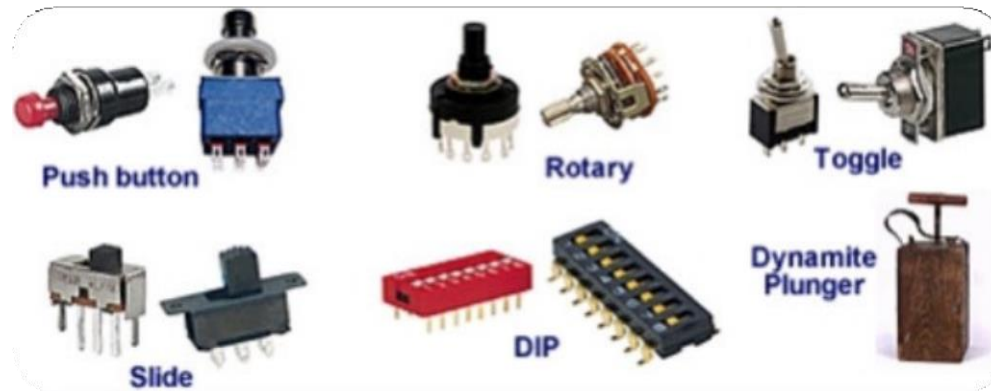


Switch **OFF**

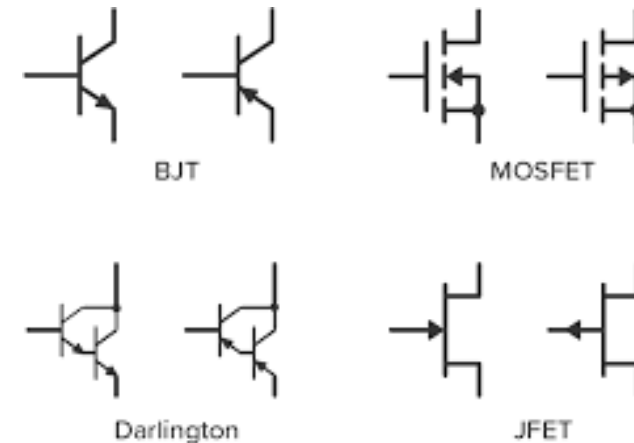


Switch – Types

- Depending on the control, the switch can be
 - **Analog:** Controlled using physical toggle/button
 - **Digital:** Controlled using voltage or current. Example – MOSFET (voltage controlled), BJT (current controlled)



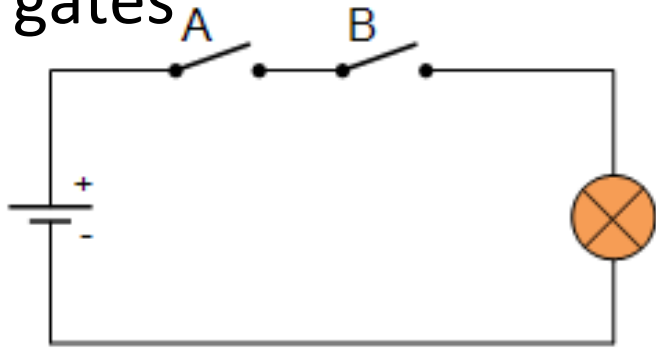
Analog switches



Digital switches
(Transistors)

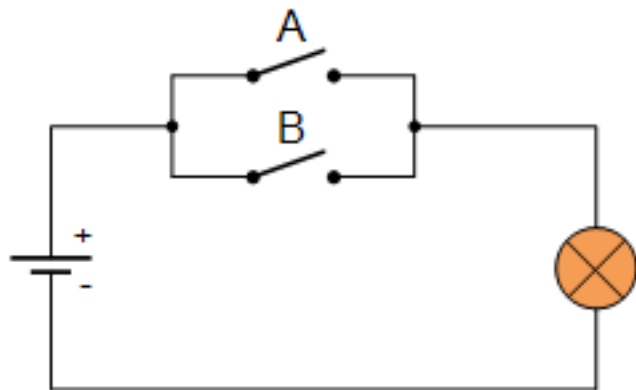
Switch Application – Logic Gates

- We can use switches to build logic gates



A	B	Bulb
0	0	OFF
0	1	OFF
1	0	OFF
1	1	ON

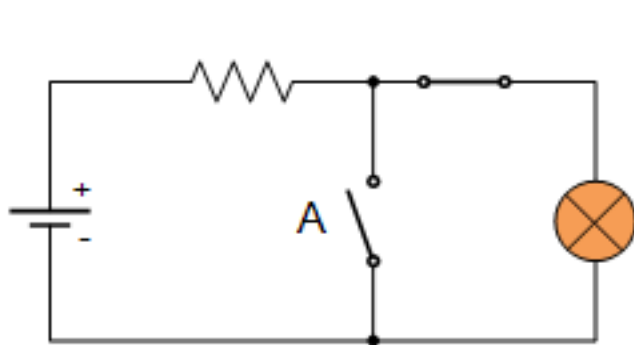
AND operation



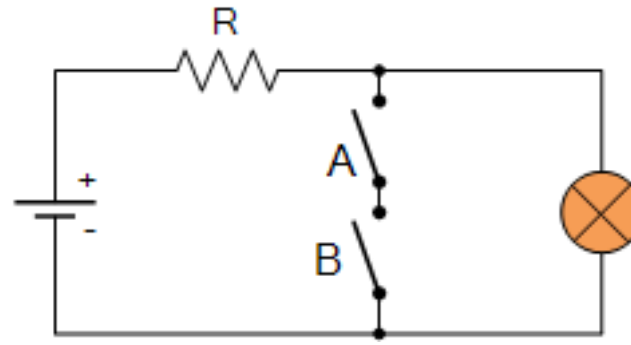
A	B	Bulb
0	0	OFF
0	1	ON
1	0	ON
1	1	ON

OR operation

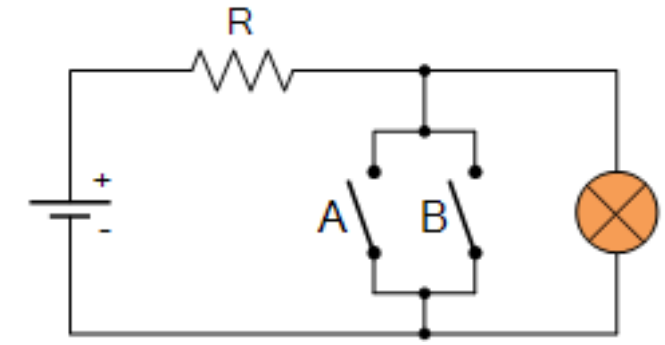
Switch Application – Logic Gates



A	Bulb
0	ON
1	OFF



A	B	Bulb
0	0	ON
0	1	ON
1	0	ON
1	1	OFF



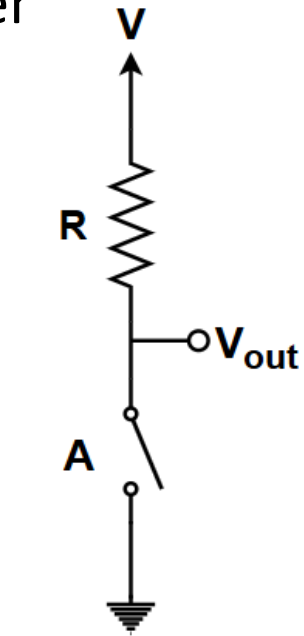
A	B	Bulb
0	0	ON
0	1	OFF
1	0	OFF
1	1	OFF

NAND operation NOR operation

These circuits are “preferred” – because they can be cascaded to build combinational logic circuits
-> if we remove the bulb and use the voltage across instead to cascade and drive the next gate

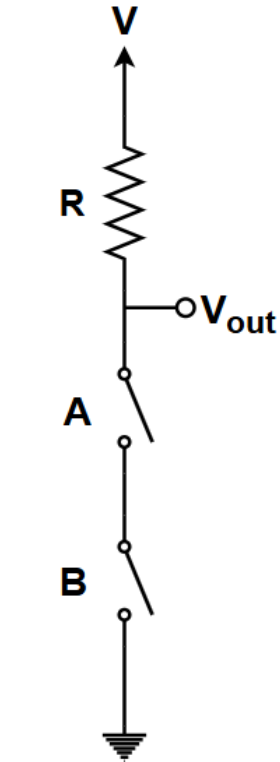
Switch Application – Logic Gates

Alternative
represer



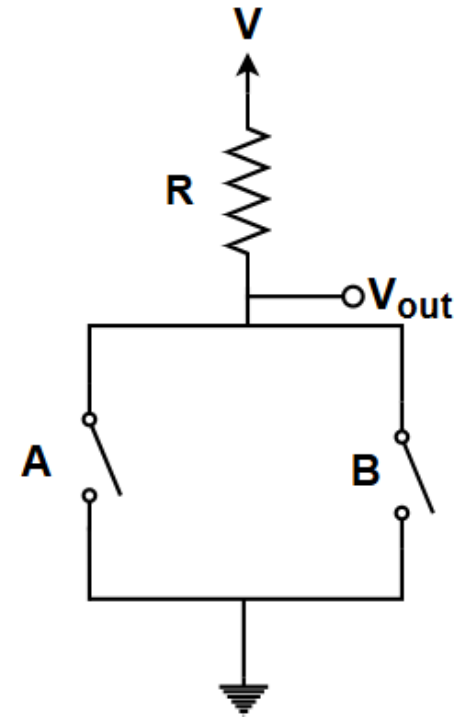
$$V_{out} = \overline{A}$$

NOT



$$V_{out} = \overline{AB}$$

NAND

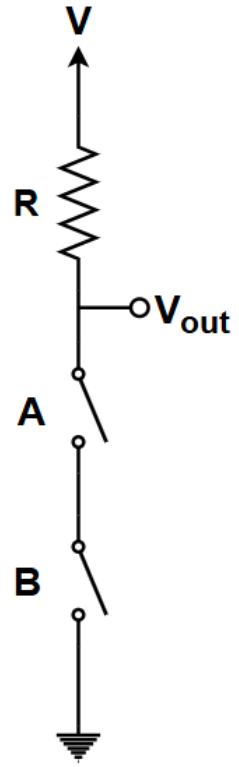


$$V_{out} = \overline{A + B}$$

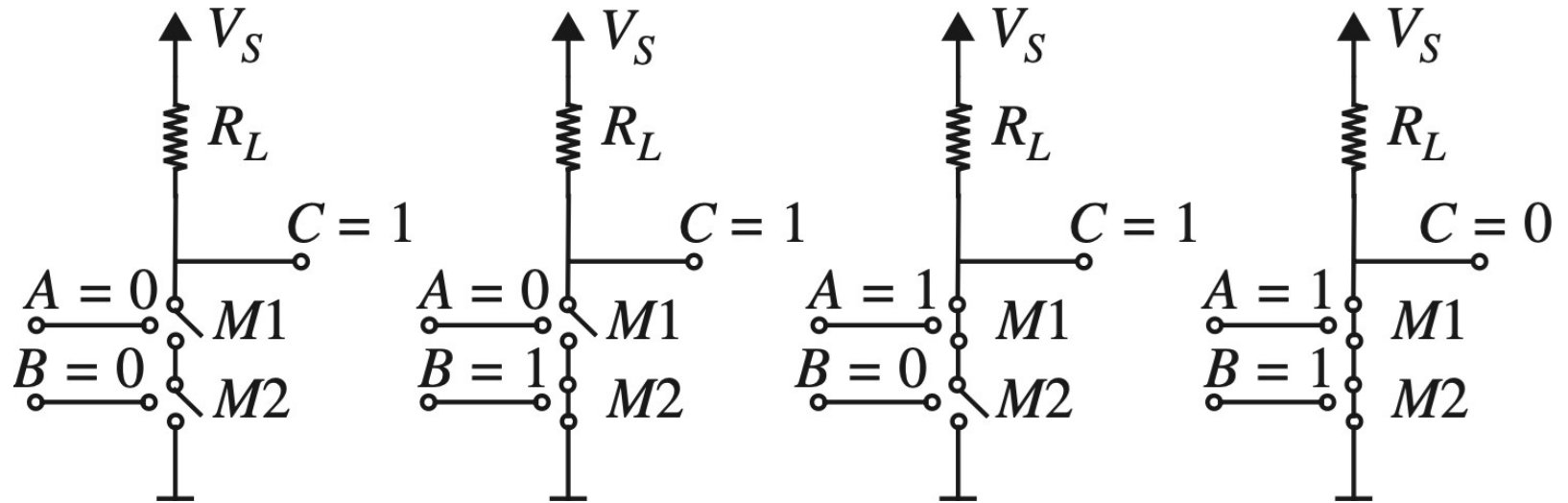
NOR

Switch Application – Logic Gates

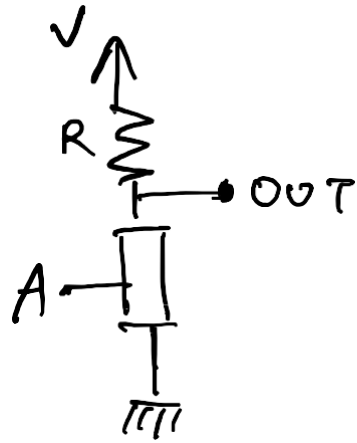
Alternative
representations:



$$V_{out} = \overline{AB}$$

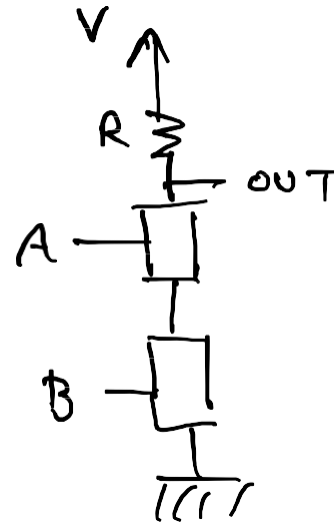


Switch Application – Logic Gates



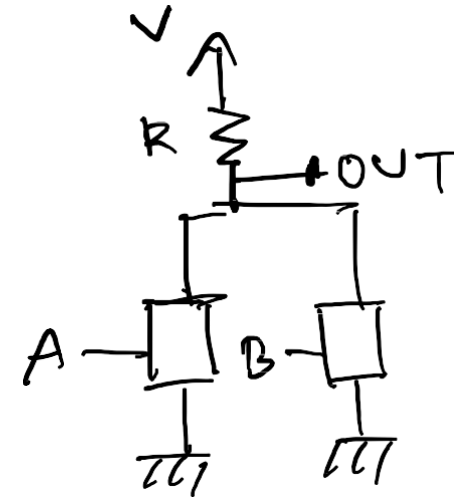
$$OUT = \overline{A}$$

A	V _{OUT}
0	5V
1	0V



$$OUT = \overline{AB}$$

A	B	V _{OUT}
0	0	5V
0	1	5V
1	0	5V
1	1	0V

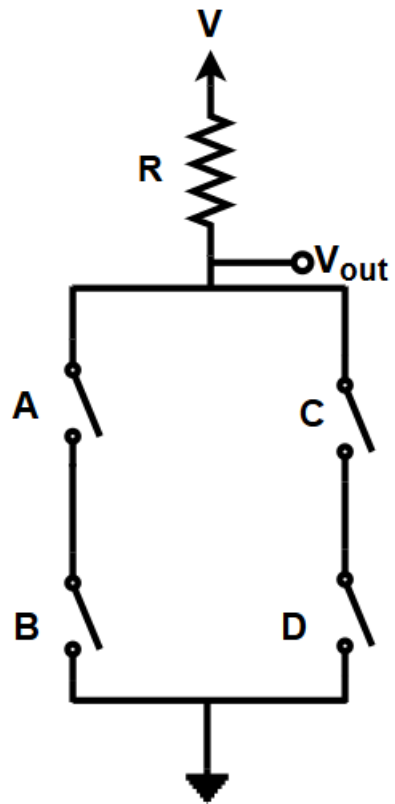


$$OUT = \overline{A+B}$$

A	B	V _{OUT}
0	0	5V
0	1	0V
1	0	0V
1	1	0V

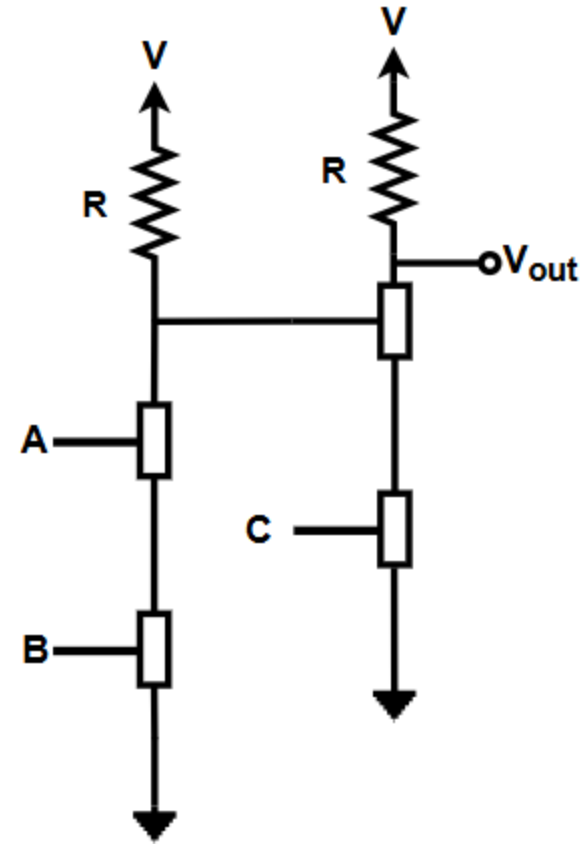
Examples

1.



$$V_{out} = \overline{AB} + \overline{CD}$$

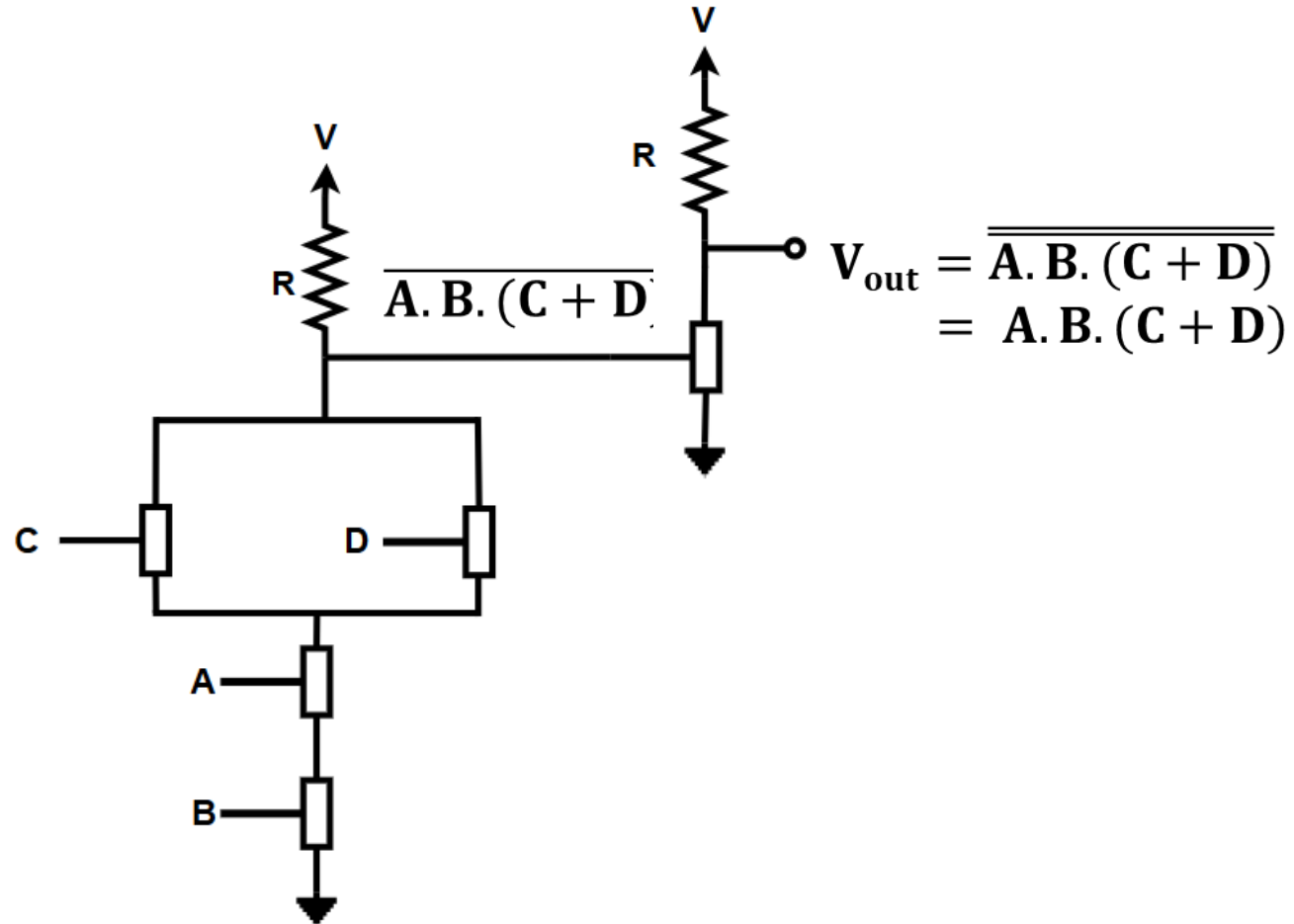
2.



$$V_{out} = \overline{\overline{AB} \cdot C}$$

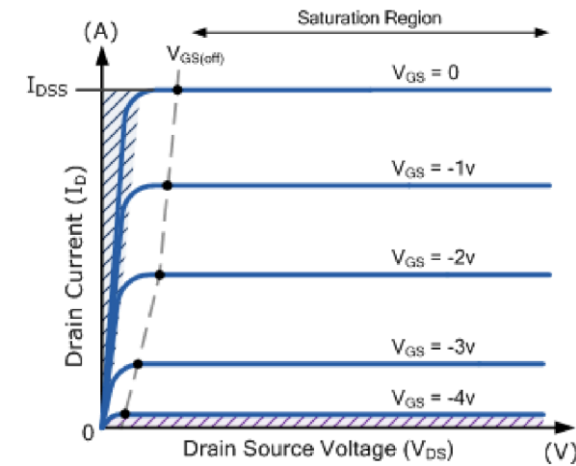
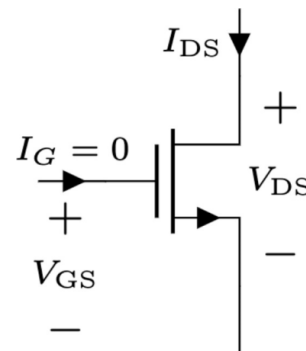
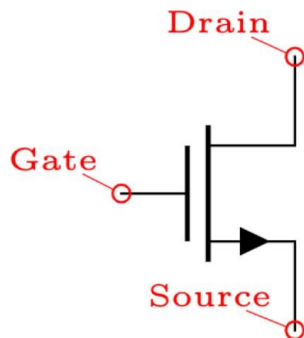
Example

Implement using switches: $f = A.B.(C+D)$



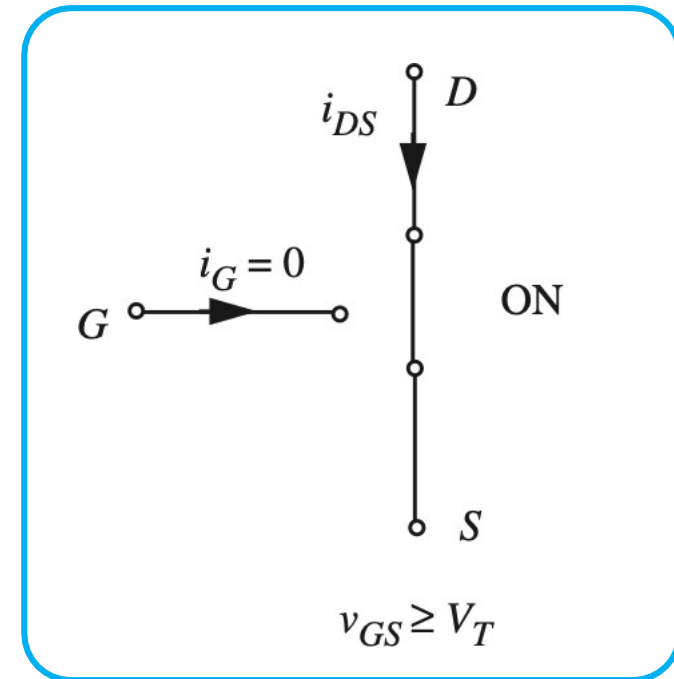
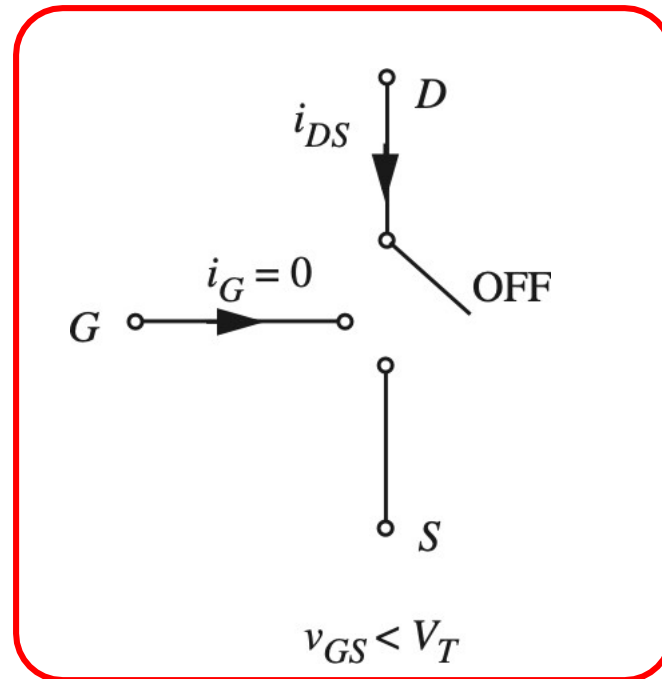
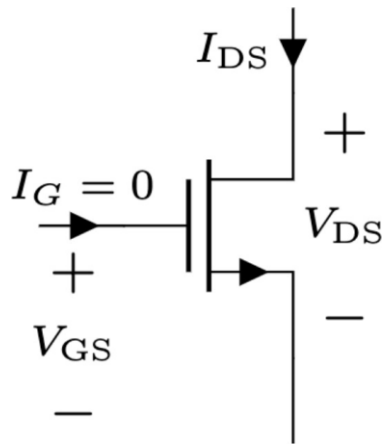
Transistors as Digital Switch

- Transistors are 3 terminal non-linear devices, can be used as switch
- 2 types – **Voltage Controlled**, **Current Controlled**
- **M**etal **O**xide **S**emiconductor **F**ield **E**ffect **T**ransistor (**MOSFET**) are **voltage controlled**
- Control, $C = V_{GS}$. The IV characteristics (I_{DS} vs V_{DS}) depends on V_{GS}
- Actual dependency is complex.
- Will start with a simple (but approximate) one – **S-Model** (Switch Model)

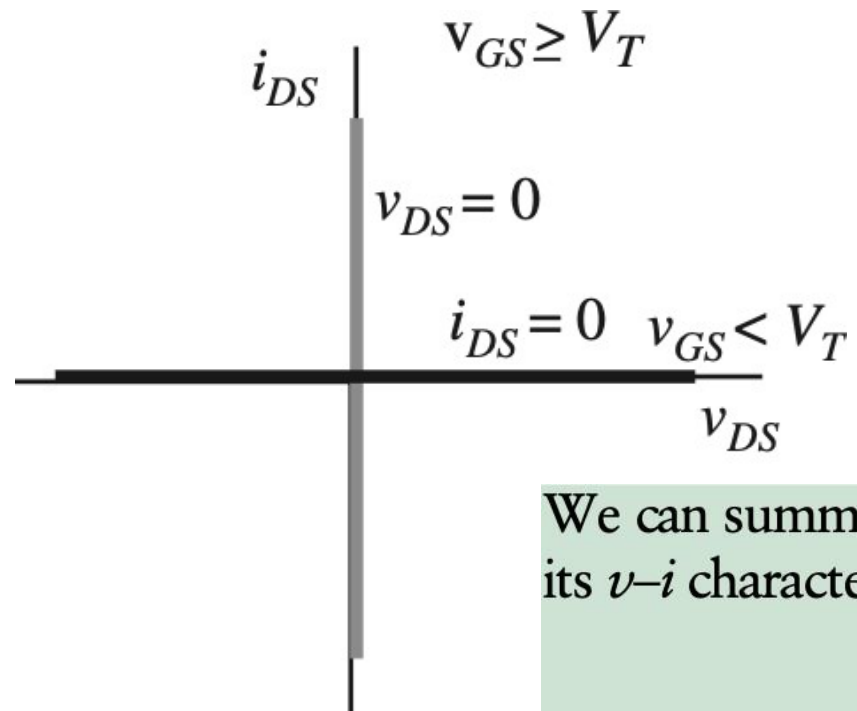


MOSFET S-Model

- The MOSFET (approximately) behaves like a switch
- $C = V_{GS}$. Here, $C = \text{"0"} \Rightarrow V_{GS} < V_T$, and $C = \text{"1"} \Rightarrow V_{GS} > V_T$



MOSFET S-Model



We can summarize the S model for the MOSFET in algebraic form by stating its v - i characteristics as follows:

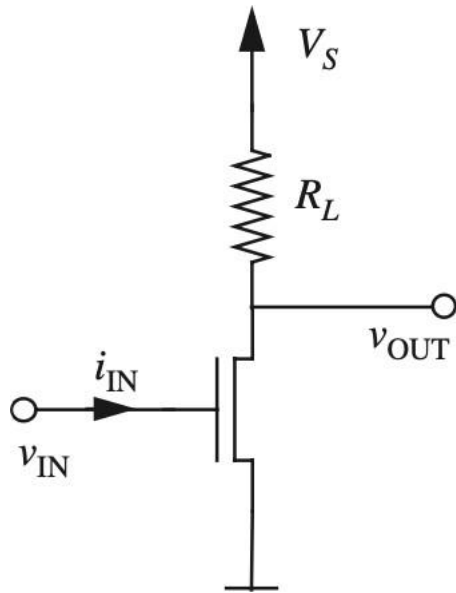
$$\text{for } v_{GS} < V_T, \quad i_{DS} = 0$$

and

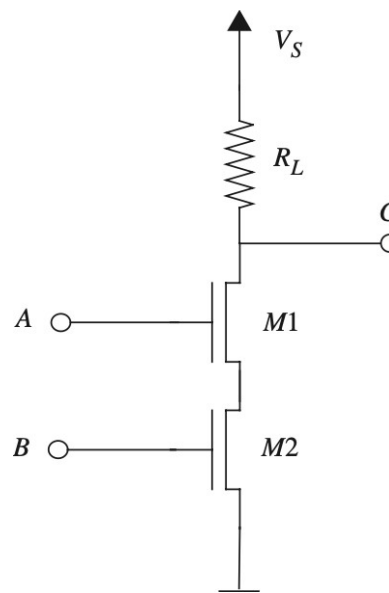
$$\text{for } v_{GS} \geq V_T, \quad v_{DS} = 0 \quad (6.2)$$

Logic Gates using MOSFET

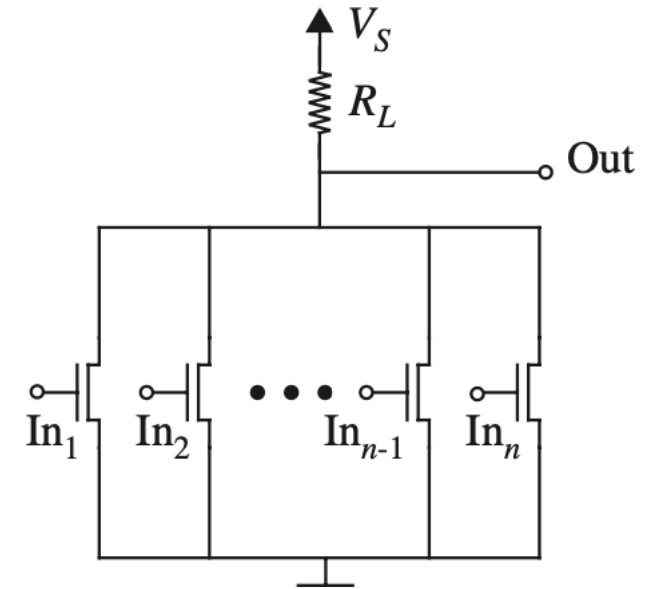
Just replace the switches with MOSFETs!



NOT Gate
(Inverter)

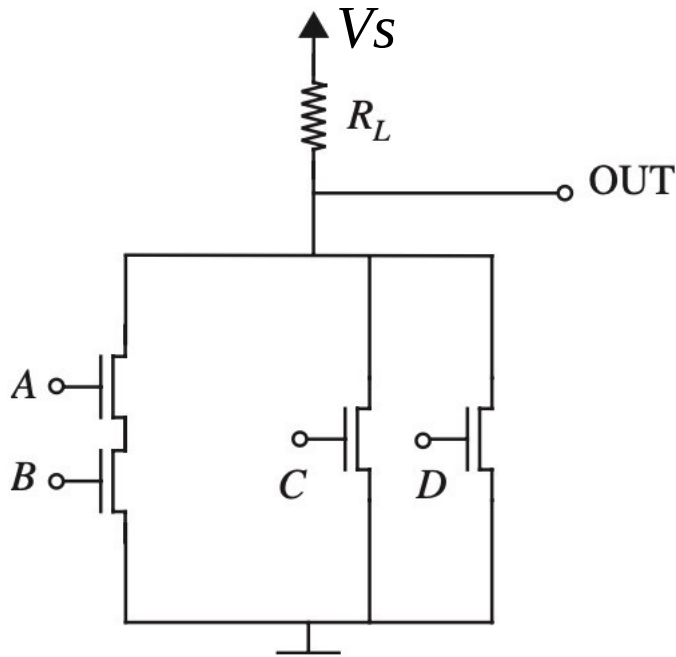


NAND Gate
(Inverter)

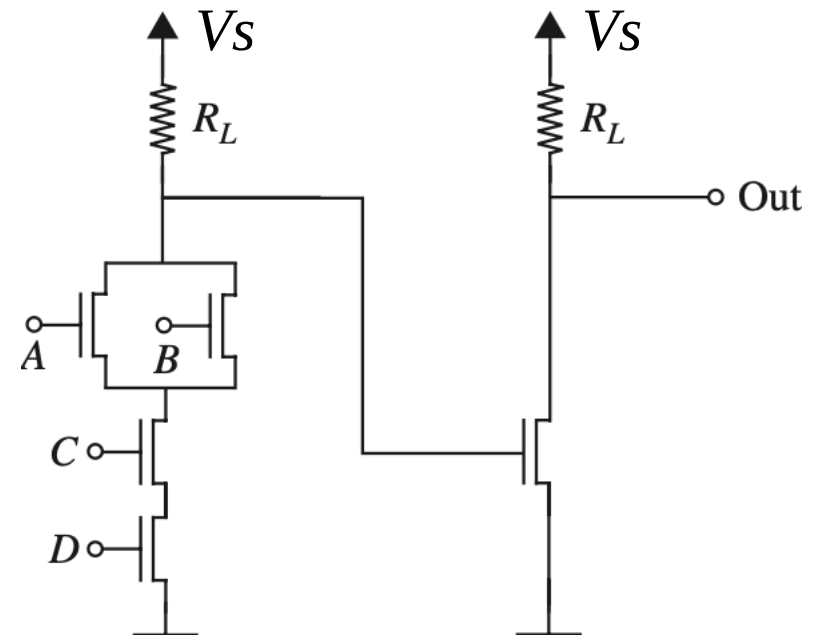


NOR Gate
(Inverter)

MOSFET Logic Gates – More Examples



$$OUT = \overline{AB + C + D}$$



$$Out = \overline{\overline{(A + B)CD}} = (A + B)CD$$

Practice Problem 2

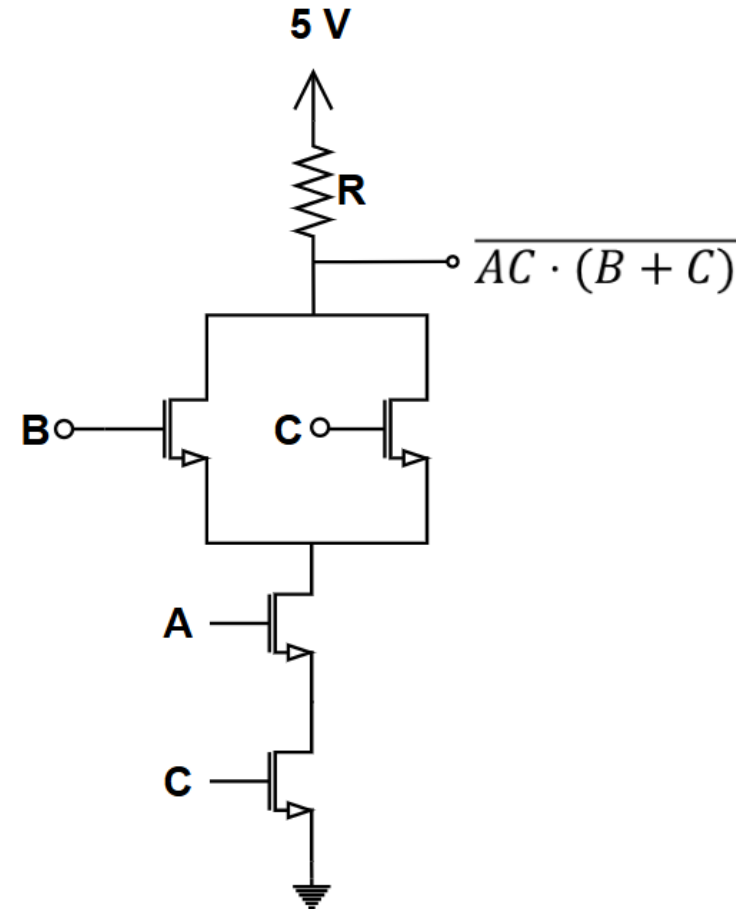
- **Design** a circuit using ideal MOSFETs (S-model) to implement the logic function

$$f = \overline{AC} + \overline{(B + C)}$$

$$= \overline{\overline{AC} + (B + C)}$$

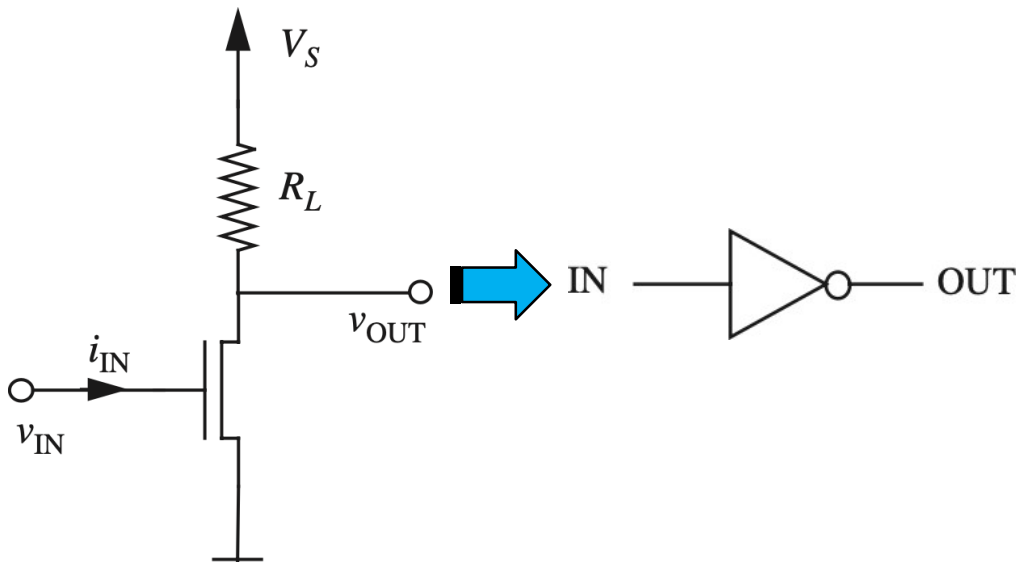
$$= \overline{\overline{AC}} \cdot \overline{\overline{(B + C)}} \quad [\text{De Morgan's Theorem}]$$

$$= \overline{AC} \cdot (B + C)$$



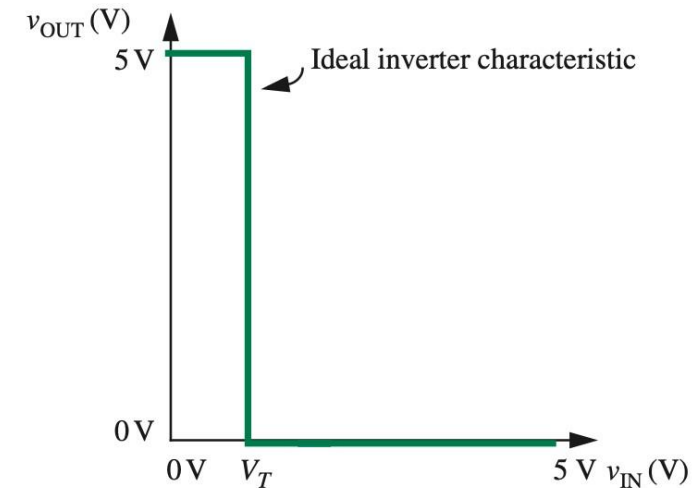
Voltage Transfer Characteristics (VTC)

- **Reminder:** VTC is a graph where x –axis = input voltage, y -axis = output voltage
- **Why?** Design logic gates to follow a given static discipline



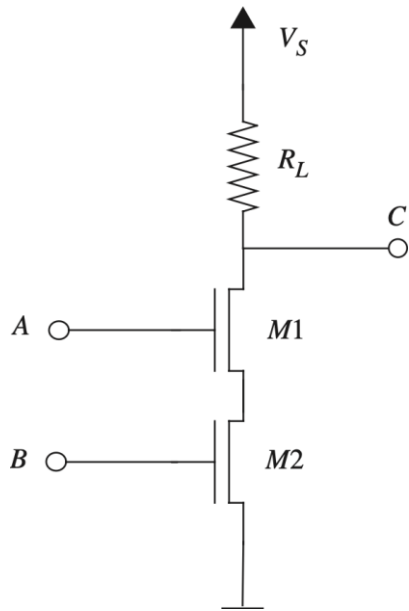
When $v_{IN} < V_T$ (Logical 0) $v_{OUT} = V_S = 5\text{ V}$ (Logical 1)

When $v_{IN} \geq V_T$ (Logical 1) $v_{OUT} = 0\text{ V}$ (Logical 0)

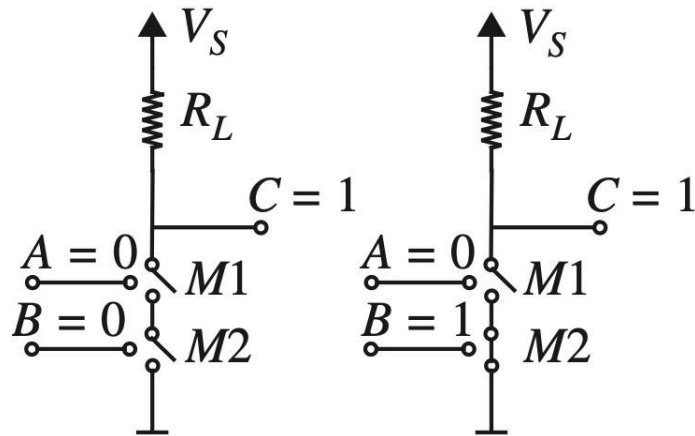


VTC of NAND gate

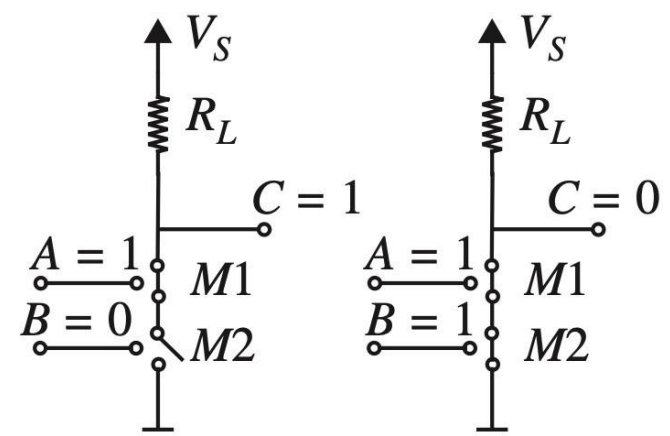
- We only have one x –axis, but two inputs
- **Solution:** Draw two VTC, one considering $V_A = 0$ one considering $V_A = 1$



When $V_A = 0$

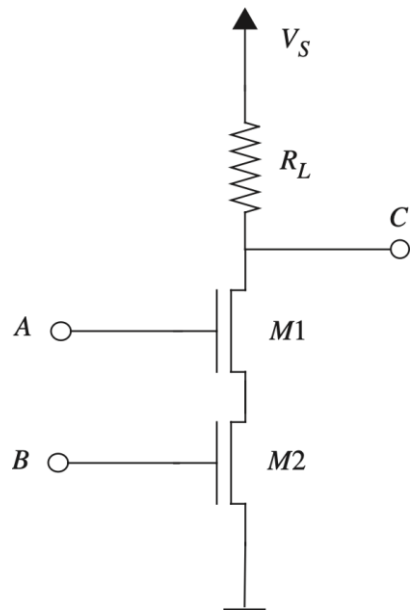


When $V_A = 1$

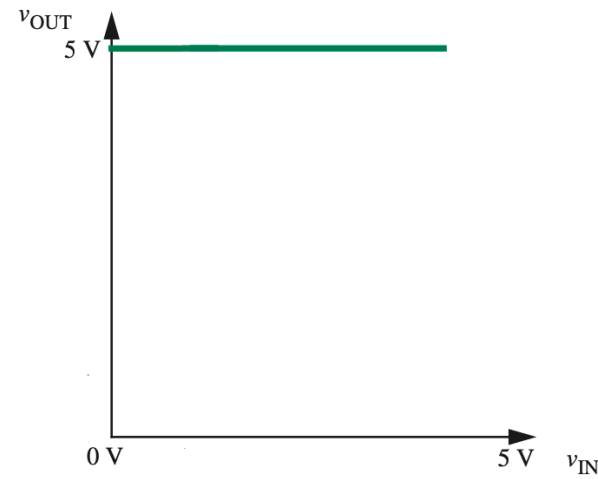


VTC of NAND gate

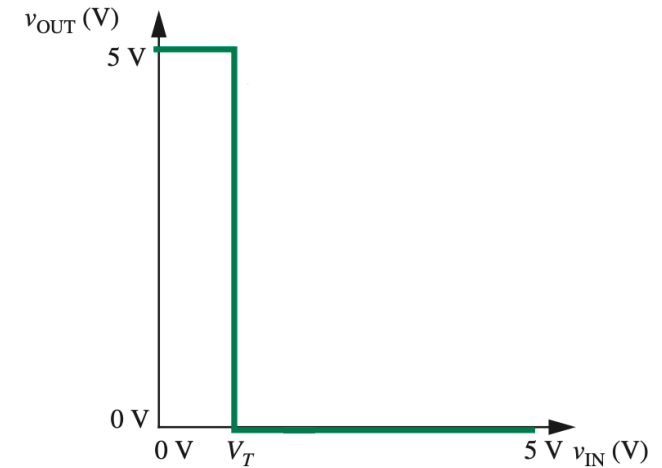
- We only have one x –axis, but two inputs
- **Solution:** Draw two VTC, one considering $V_A = 0$ one considering $V_A = 1$



When $V_A = 0$



When $V_A = 1$



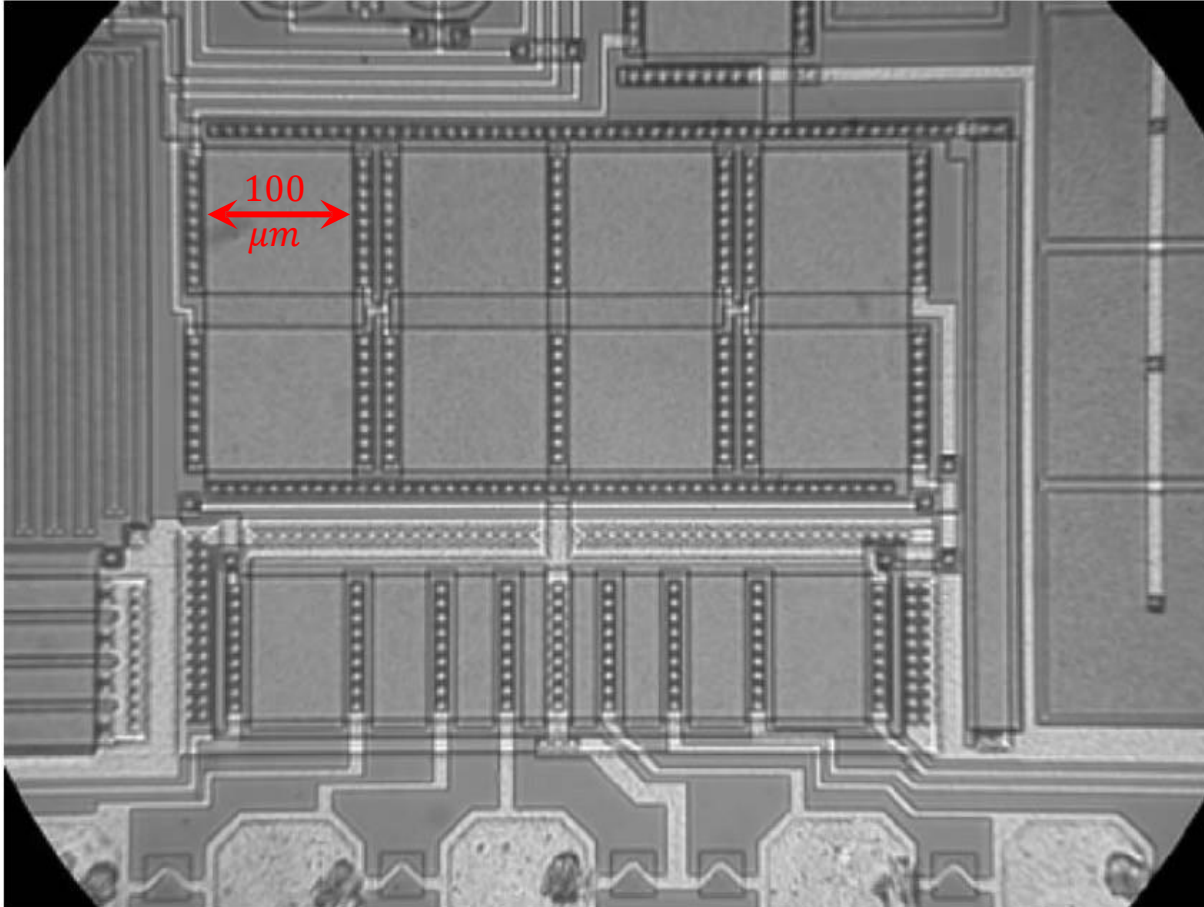
Homework: Find VTC for NOR

gate

Outline

- Constructing a *real* MOSFET – n/p-channel
- **Operation of an MOSFET-**
 1. Cut-Off
 2. Saturation
 3. Triode Mode
- Output Characteristics
- PWL Model and Non-ideal Analysis: **SR model**
- Real MOSFET equations
- Introduction to Static analysis

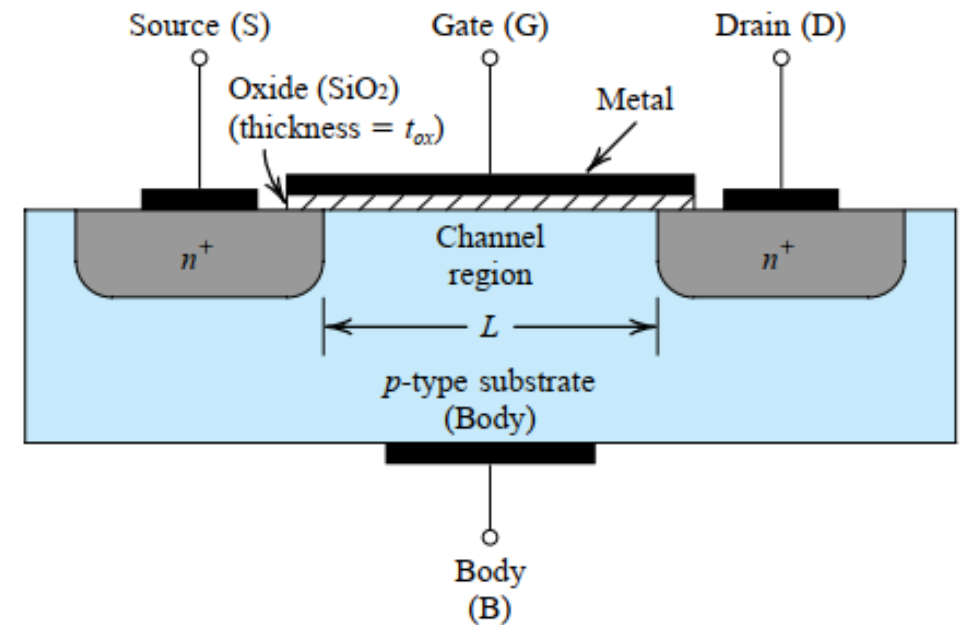
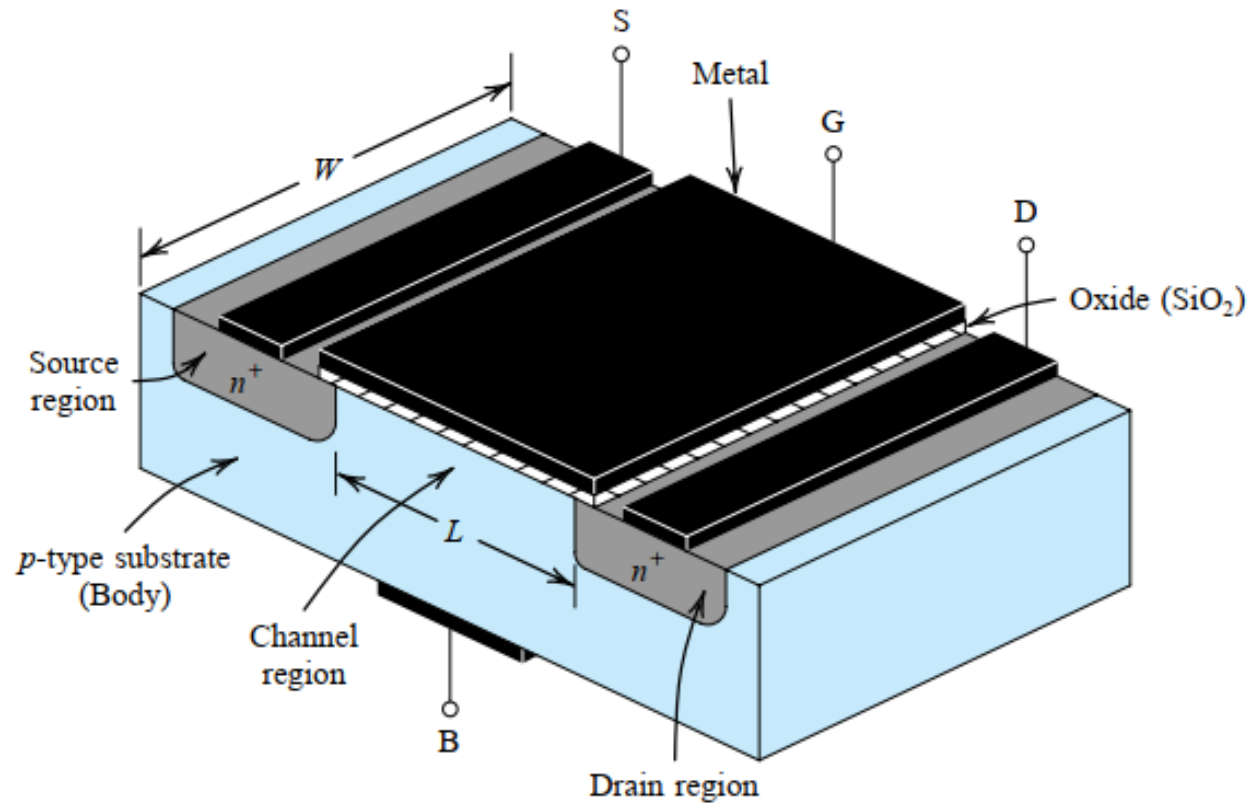
Construction of Real MOSFET



Top view of several n-channel MOSFETs fabricated on a chip. The square MOSFETs in the center of the photograph have a width and length of $100\text{ }\mu\text{m}$. (Photograph Courtesy of Maxim Integrated Products.)

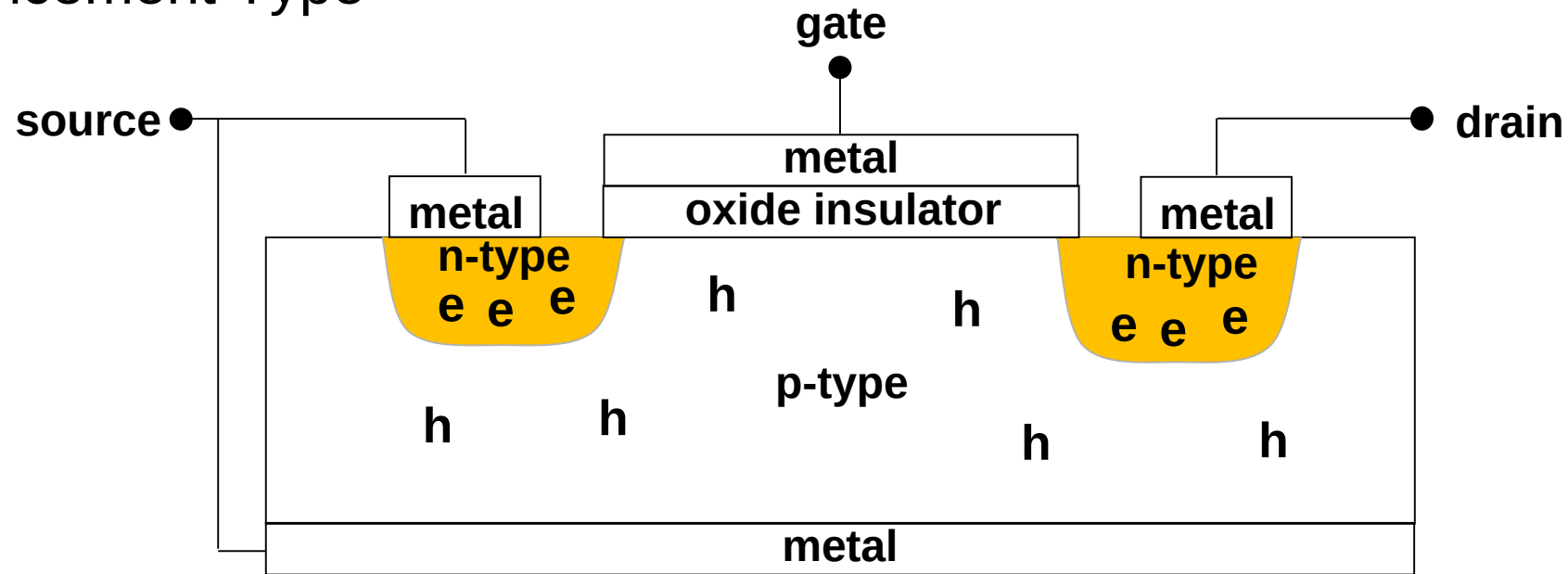
Real MOSFET – Enhancement Type

Device Structure (*n*-channel MOSFET)

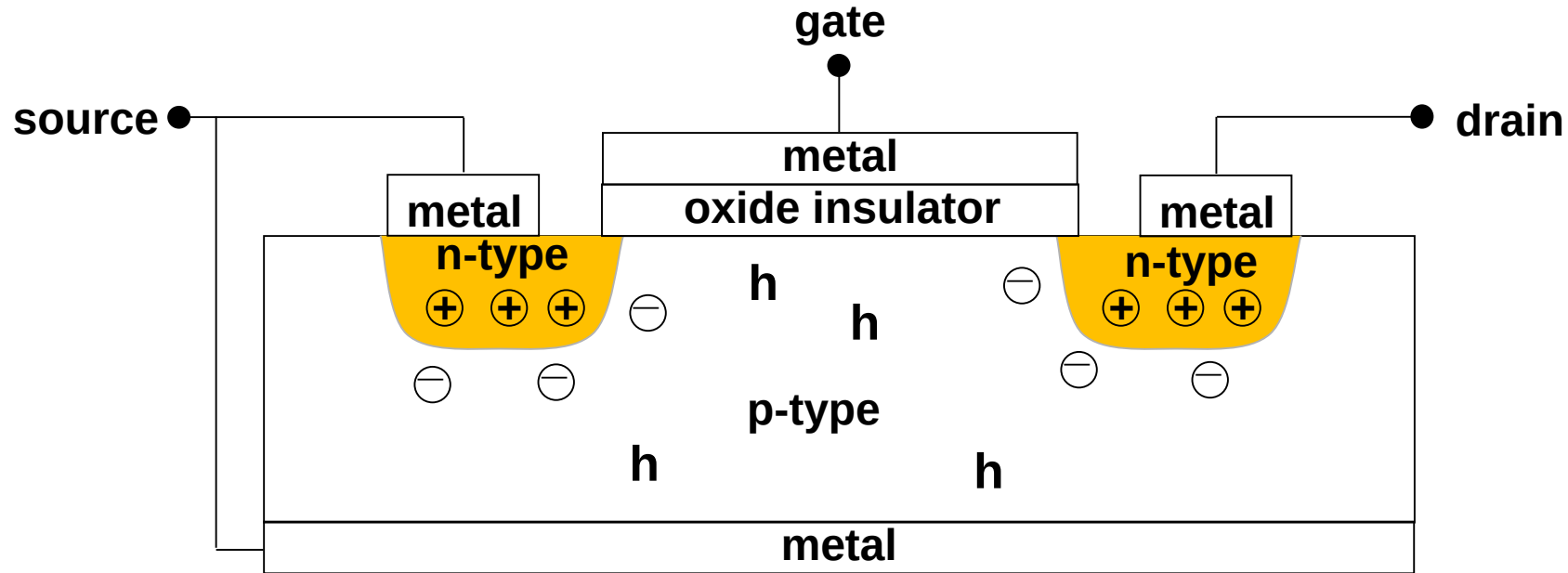


n-channel MOSFET (NMOS) Physical Structure

Enhancement Type



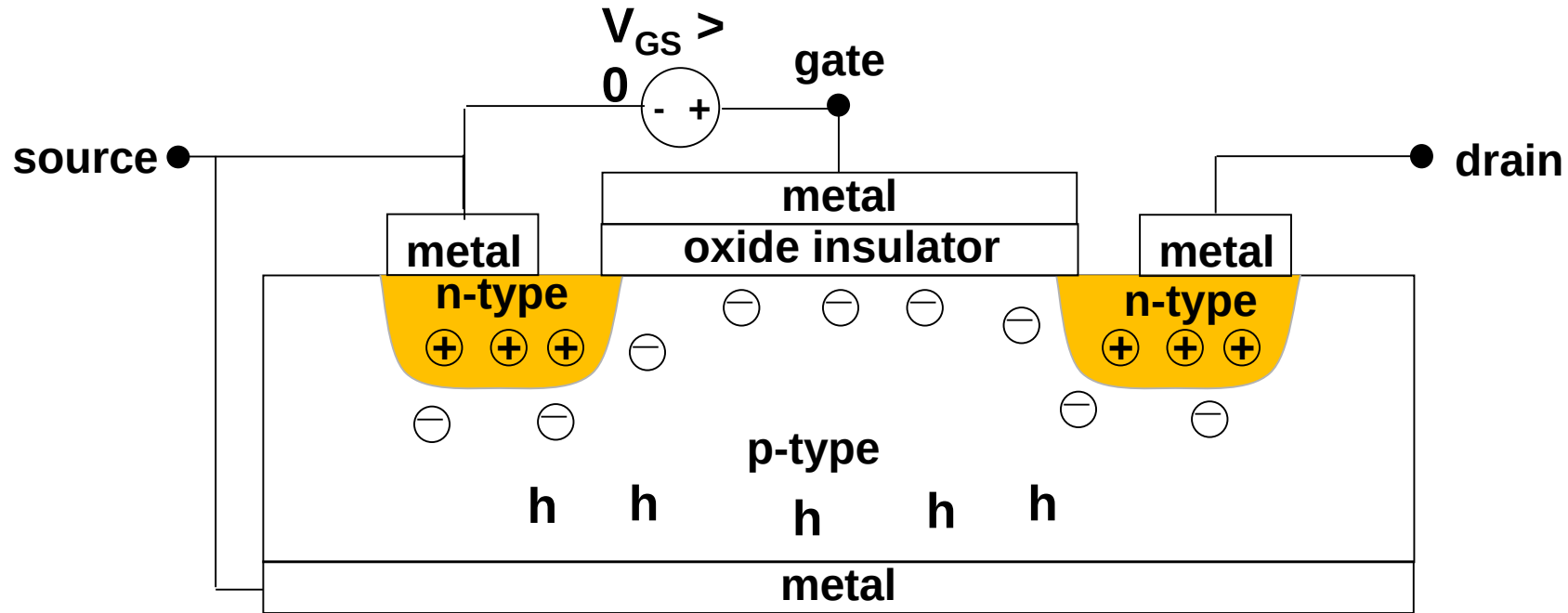
NMOS in Equilibrium



When the transistor is left alone, some electrons from the n-type wells diffuse into the p-type material to fill holes.

This creates negative ions in the p-type material and positive ions are left behind in the n-type material.

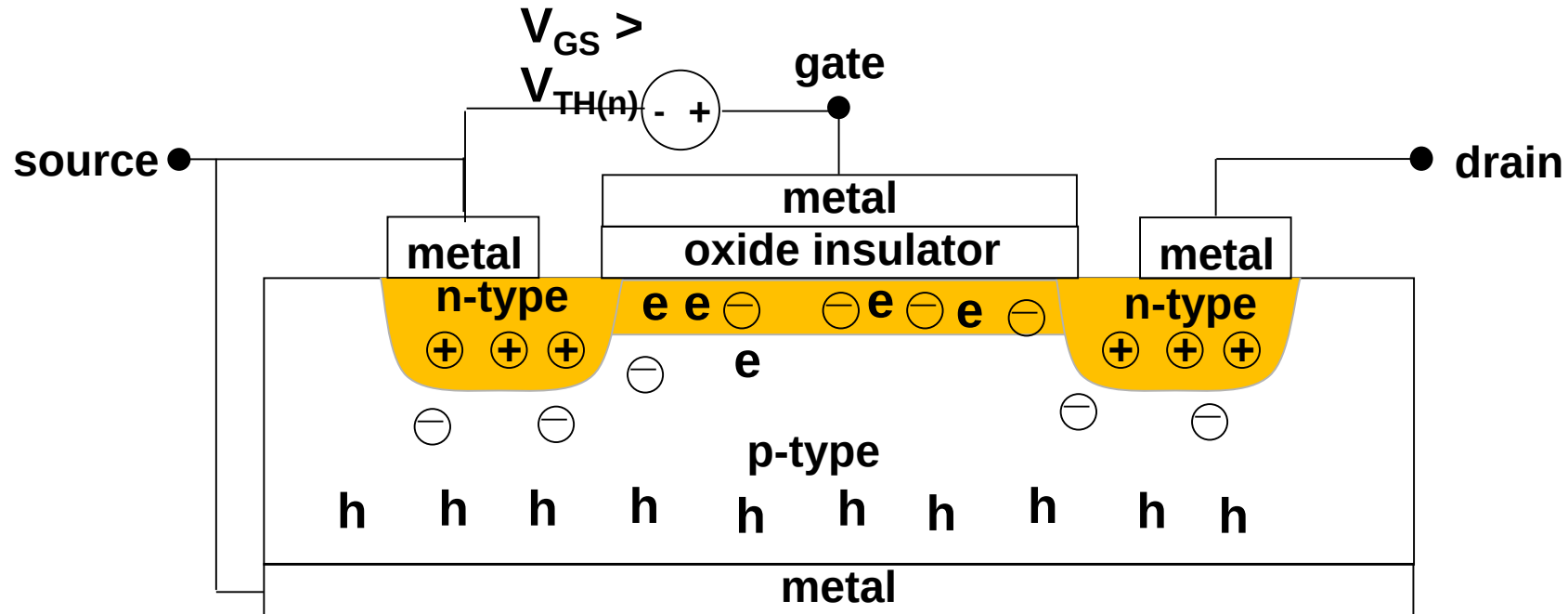
NMOS in Cutoff



When a small, positive V_{GS} is applied, holes “move away” from the gate.

Electrons from complete atoms elsewhere in the p-type material move to fill holes near the gate instead.

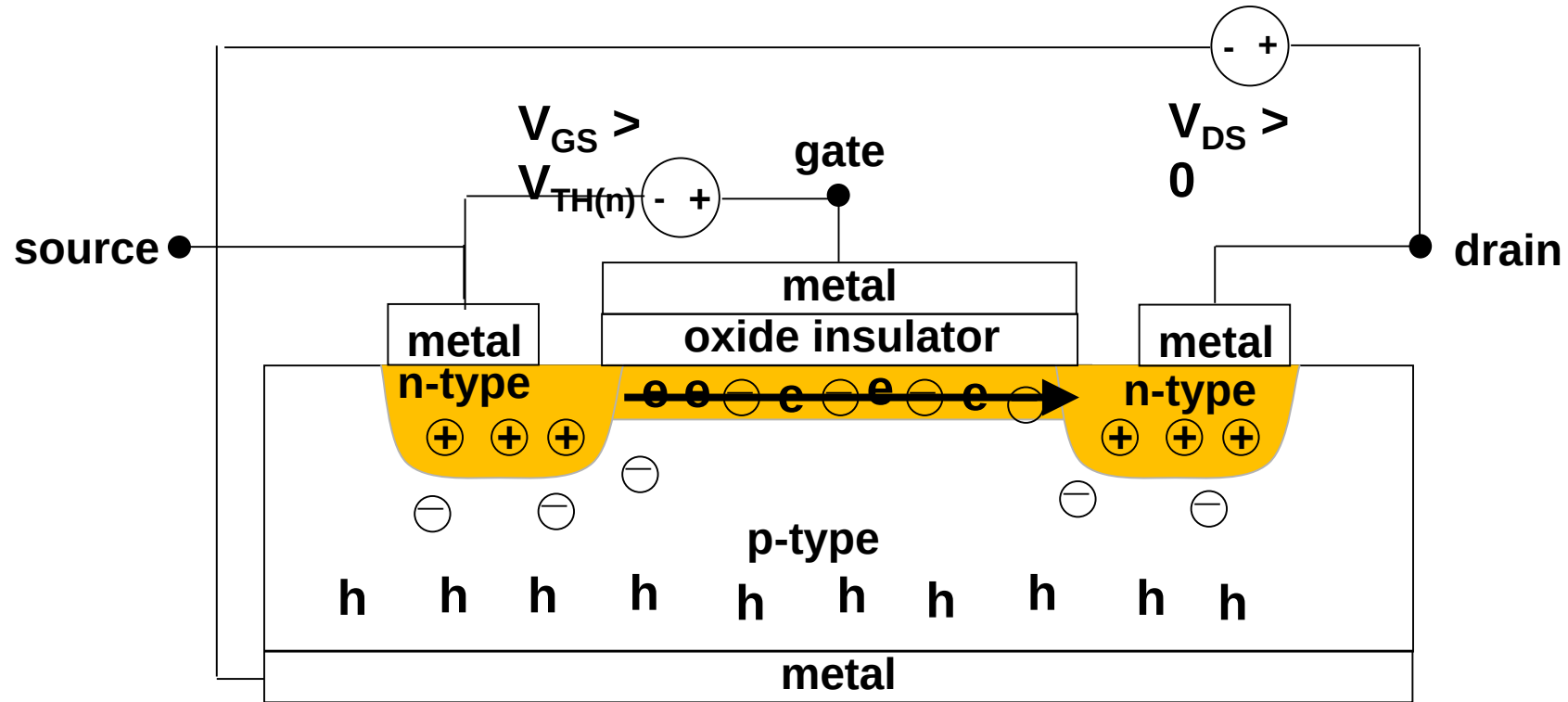
NMOS Transistor channel



When V_{GS} is larger than a **threshold** voltage $V_{TH(n)}$, the attraction to the gate is so great that free electrons collect there.

The applied V_{GS} creates an **induced n-type channel** under the gate (an area with free electrons).

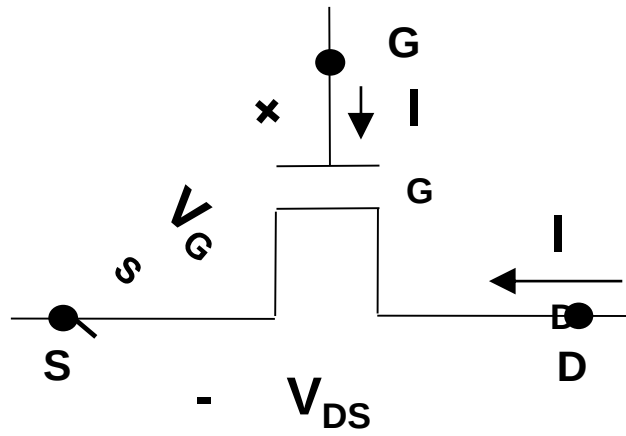
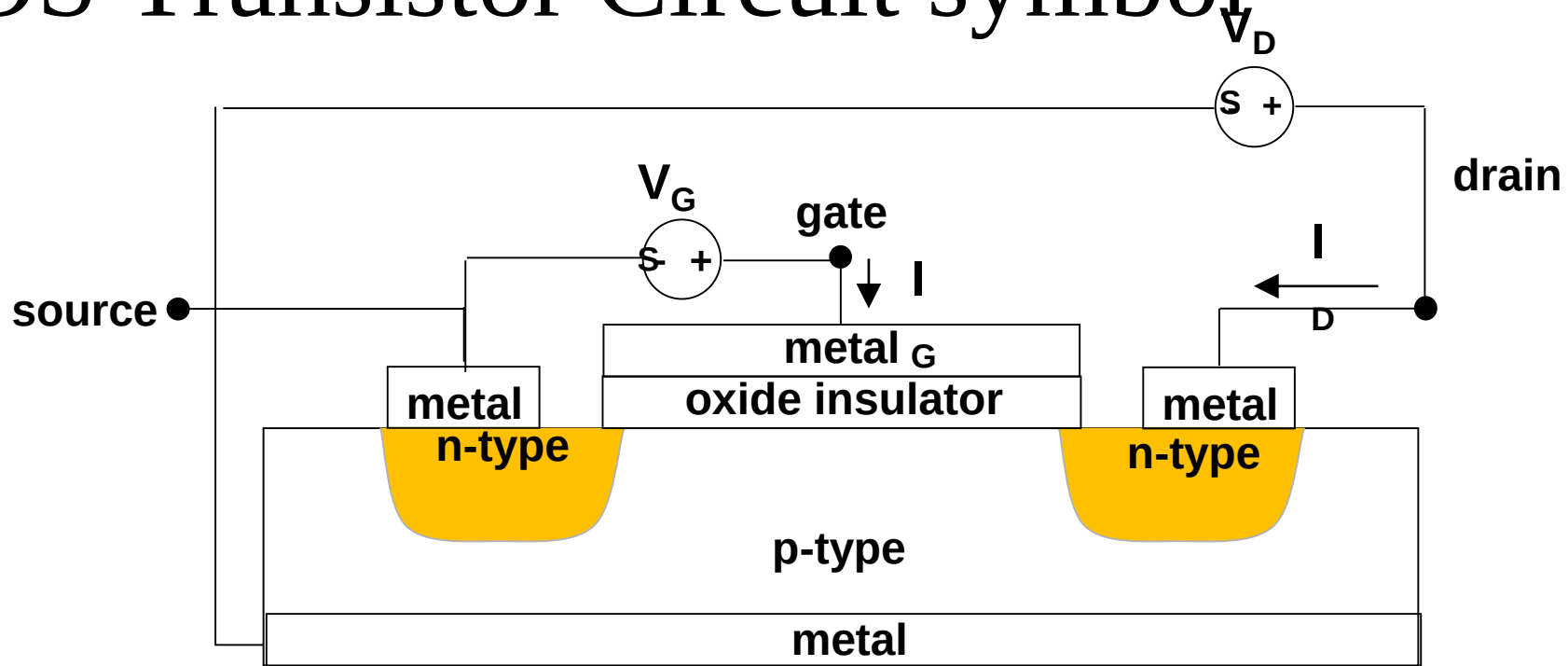
NMOS Transistor Drain Current



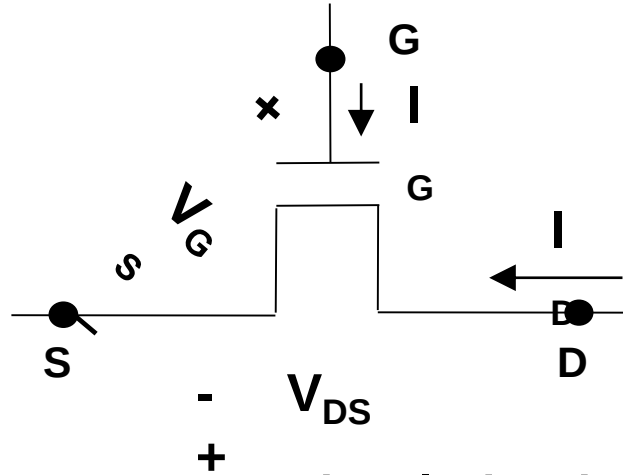
When a positive V_{DS} is applied, the free electrons flow from the source to the drain. (Positive current flows from drain to source).

The amount of current depends on V_{DS} , as well as the number of electrons in the channel, channel dimensions, and material.

NMOS Transistor Circuit symbol



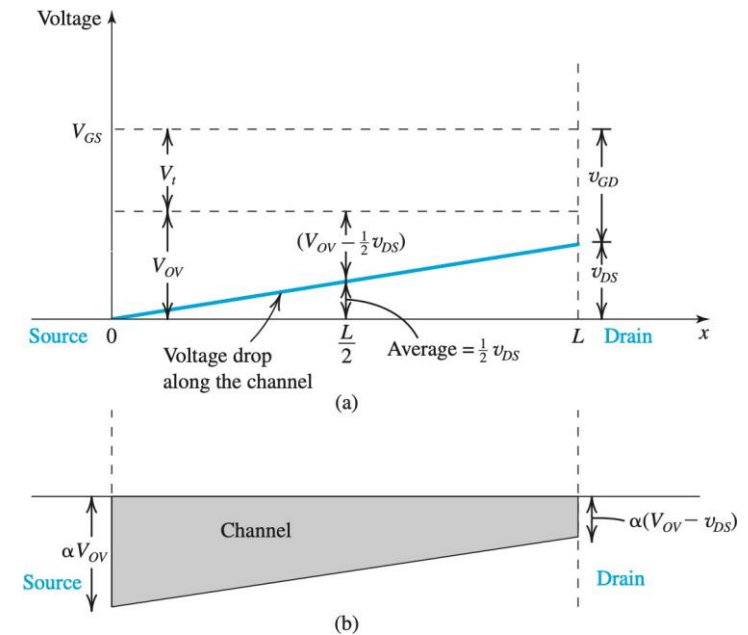
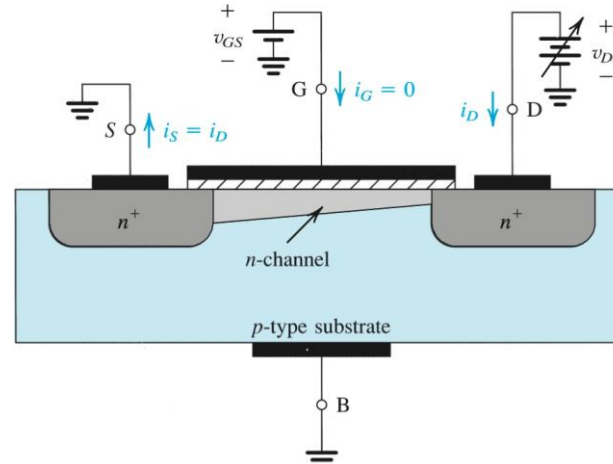
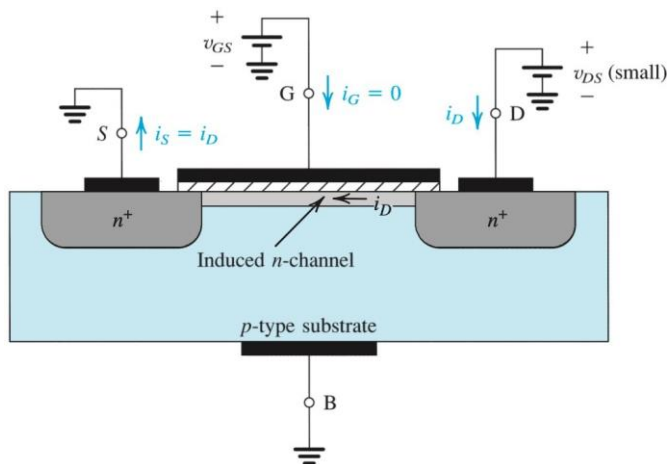
NMOS I-V Characteristic



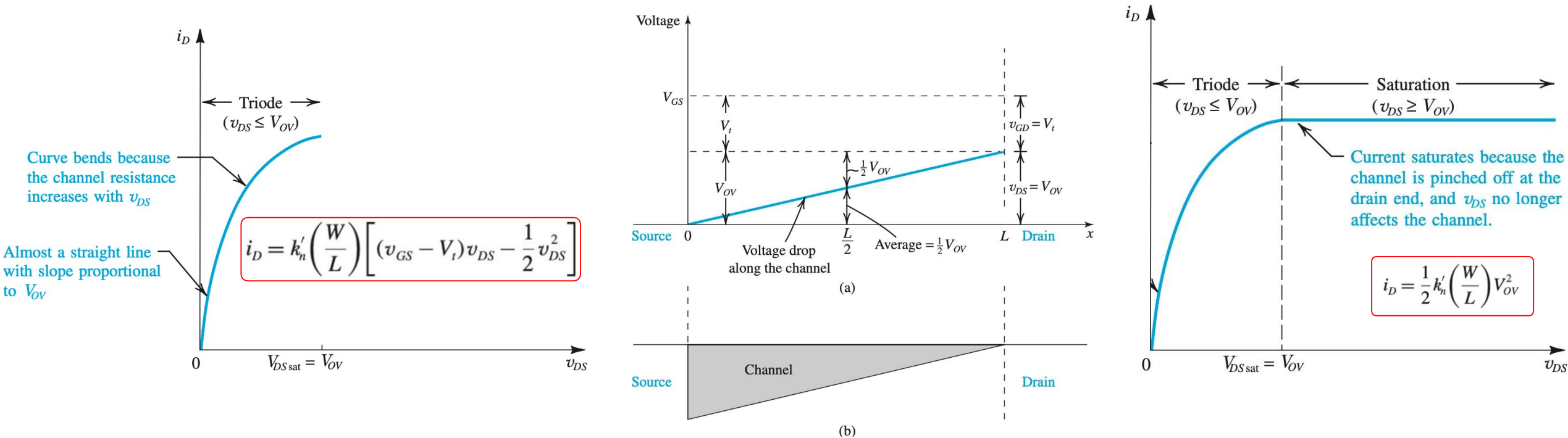
- Since the transistor is a 3-terminal device, there is no single I-V characteristic.
- Note that because of the insulator, $I_G = 0$ A.
- We typically define the MOS I-V characteristic as I_D vs. V_{DS} for a fixed V_{GS} .
- The I-V characteristic changes as V_{GS} changes.

Real MOSFET

- For small V_{DS} , uniform channel, hence fixed R_{ON} therefore SR model valid.
- As V_{DS} is increased, channel becomes tapered cause $V_{GD} \downarrow$. Resistance $\uparrow$, slope $\downarrow$.
- This mode is called the **triode mode**. Condition: $V_{DS} < V_{OV}$



Real MOSFET



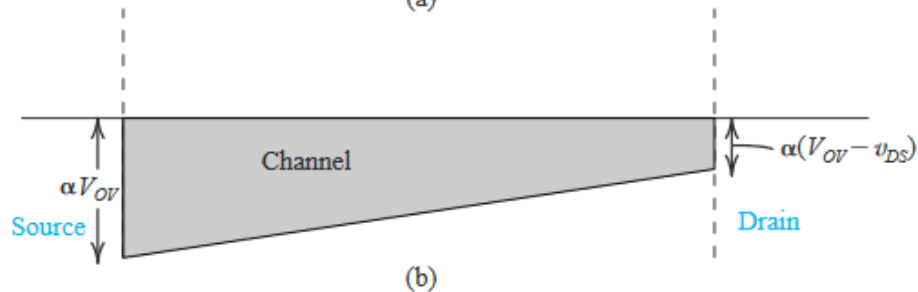
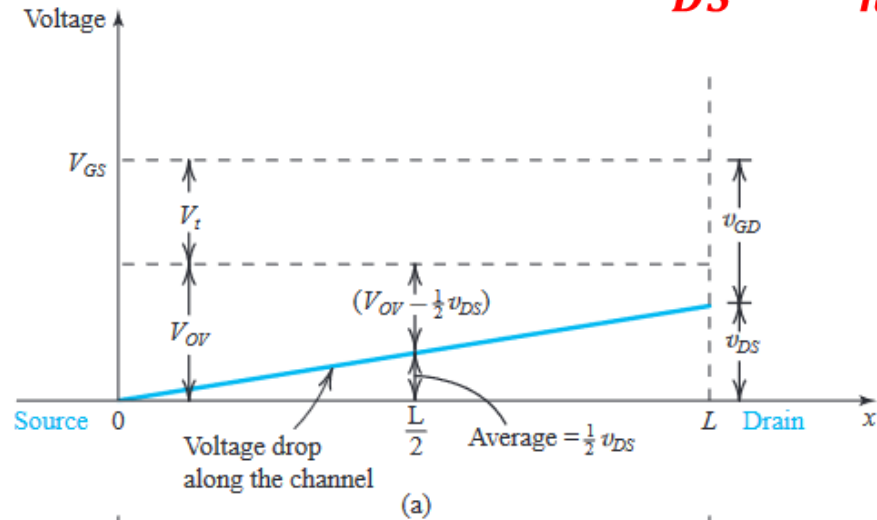
- When $V_{DS} = V_{OV}$, channel pinches off.
- Increasing V_{DS} further have no effect on channel shape. Hence, current saturates
- This mode is called the **saturation mode**. Condition: $V_{DS} \geq V_{OV}$
- Behaves like a current source (constant current) that depends on V_{OV} .

NMOS Triode Mode

Occurs when $v_{GS} \geq V_T$ and $v_{DS} < v_{GS} - V_T$

$$I_{DS} = k_n \left(v_{GS} - V_T - \frac{1}{2} v_{DS} \right) v_{DS}$$

$$\begin{aligned} I_{DS} &\propto k'_n \\ I_{DS} &\propto \frac{W}{L} \\ I_{DS} &\propto (V_{OV} - \frac{1}{2} v_{DS}) \end{aligned}$$



Overdrive Voltage is defined as:

$$V_{OV} = v_{GS} - V_T$$

In triode mode, average voltage across channel is actually:

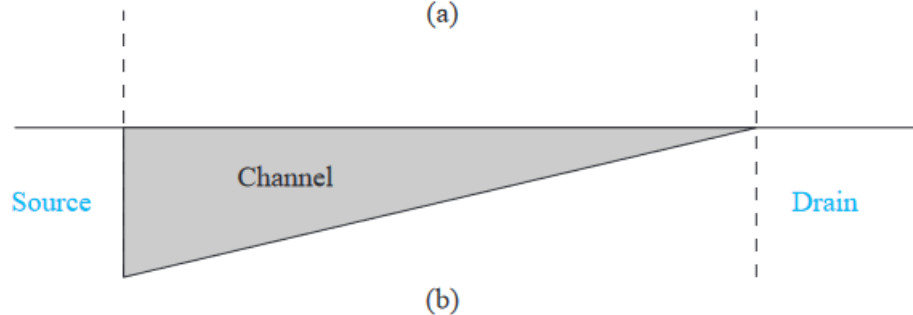
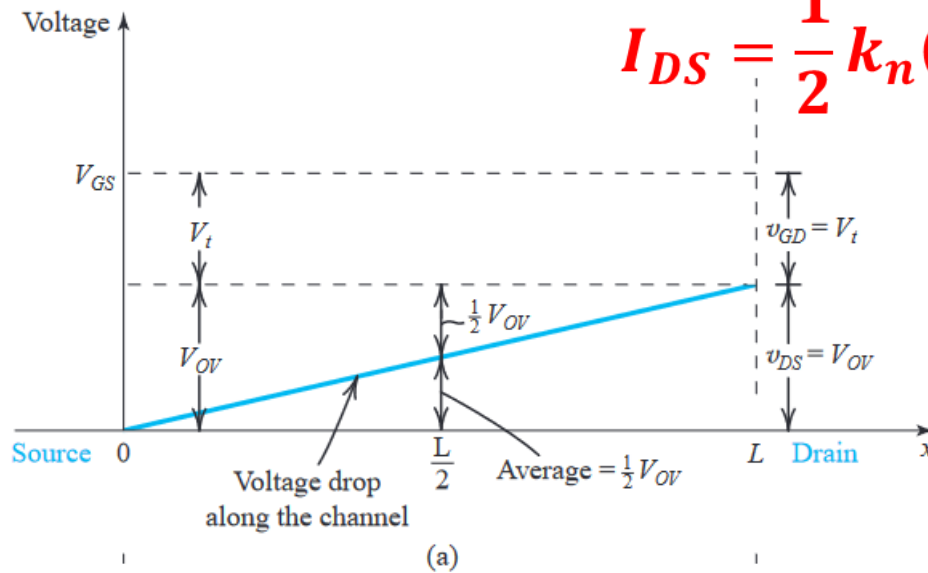
$$(V_{OV} - \frac{1}{2} v_{DS})$$

NMOS Saturation mode

Occurs when $v_{GS} > V_T$ and $v_{DS} \geq v_{GS} - V_T$

$$I_{DS} = \frac{1}{2} k_n (v_{GS} - V_T)^2 = \frac{1}{2} k_n V_{OV}^2$$

$$\begin{aligned} I_{DS} &\propto k'_n \\ I_{DS} &\propto \frac{W}{L} \end{aligned}$$



Overdrive Voltage is defined as:

$$V_{OV} = v_{GS} - V_T$$

In **Saturation** mode, average voltage across channel fixed at:

$$\frac{1}{2} V_{OV}$$

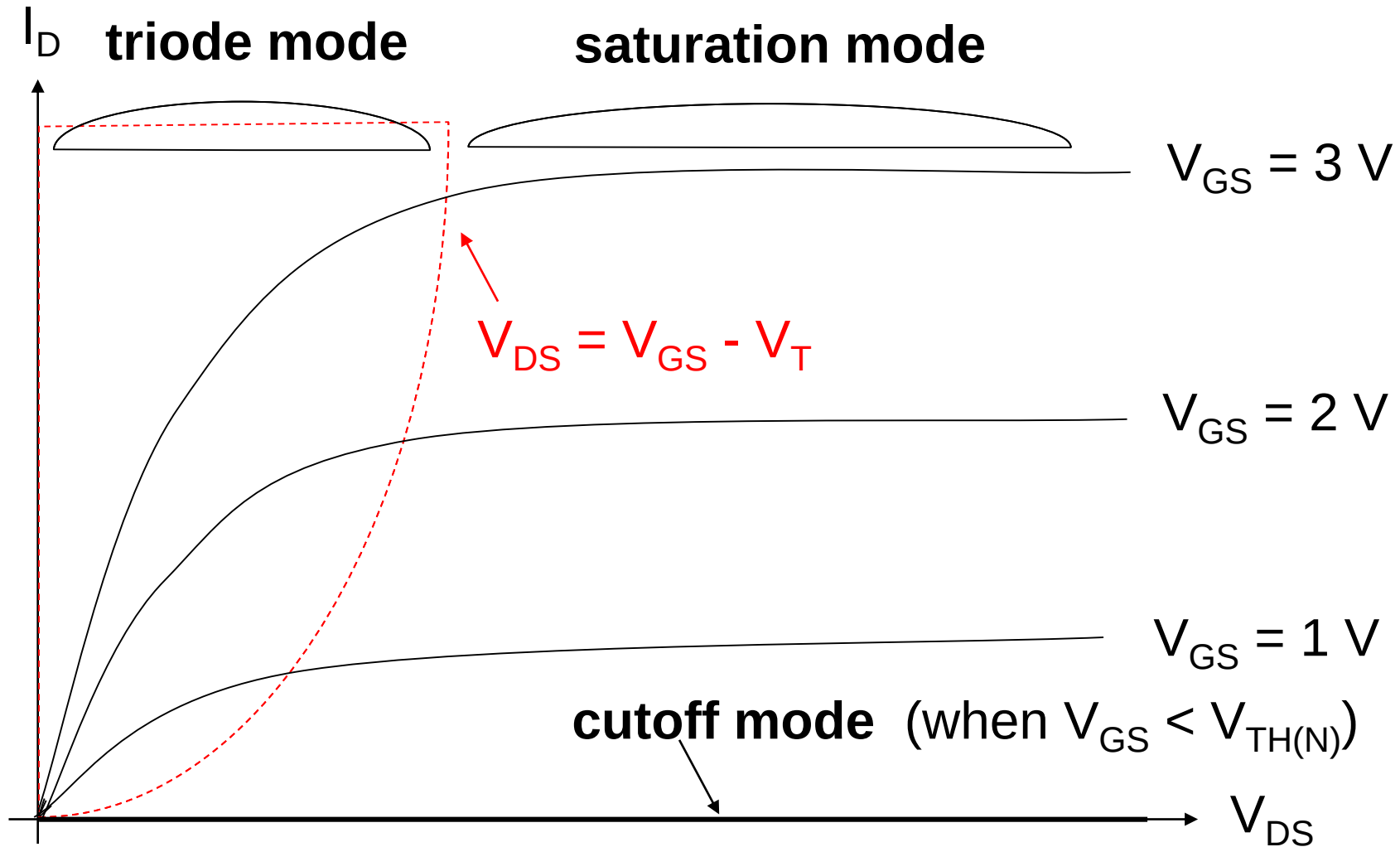
Modes of Operation

For small values of V_{GS} , $V_{GS} \leq V_{TH(n)}$, the n-type channel is not formed. No current flows. This is **cutoff mode**.

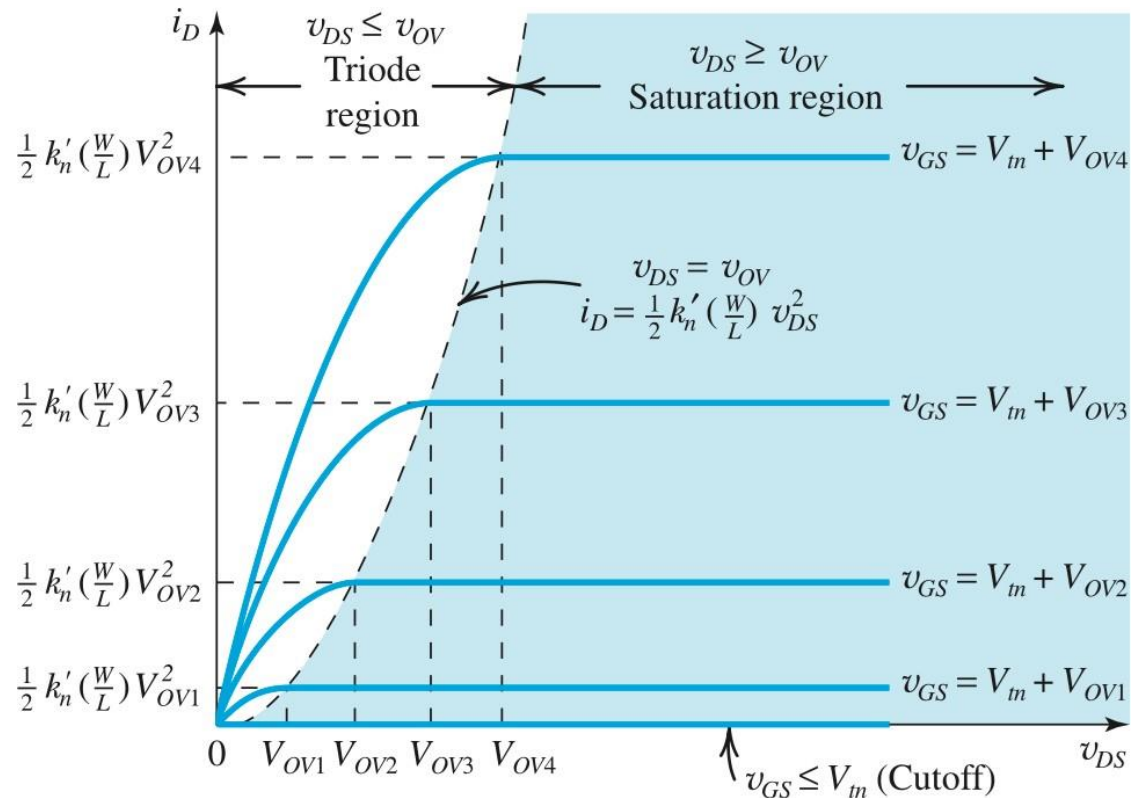
When $V_{GS} > V_{TH(n)}$, current I_D may flow from drain to source, and the following modes of current flow are possible.

- The mode of current flow depends on the propelling voltage, V_{DS} , and the channel-inducing voltage, $V_{GS} - V_{TH(n)}$.
- When $V_{DS} < V_{GS} - V_{TH(n)}$, current is starting to flow. I_D increases rapidly with increased V_{DS} . This is **triode mode**.
- When $V_{DS} \geq V_{GS} - V_{TH(n)}$, current is reaching its maximum value. I_D does not increase much with increased V_{DS} . This is called **saturation mode**.

NMOS I-V Characteristic



IV Characteristics of Real MOSFET



Mode	Condition	Equation
Cutoff	$V_{GS} < V_T$	$I_D = 0$
Triode	$V_{GS} \geq V_T$ $V_{DS} < V_{OV}$	$I_D = k[V_{OV}V_{DS} - \frac{1}{2}V_{DS}^2]$
Saturation	$V_{GS} \geq V_T$ $V_{DS} \geq V_{OV}$	$I_D = \frac{k}{2}V_{OV}^2$

$$V_{GS} - V_T = V_{OV}$$

$$k_n = \frac{k'_n W}{L}$$

Water Tap Analogy

Imagine the water tap on your kitchen sink.

To make water flow, the water supply has to be connected to the water tap. This establishes a path for water to flow.

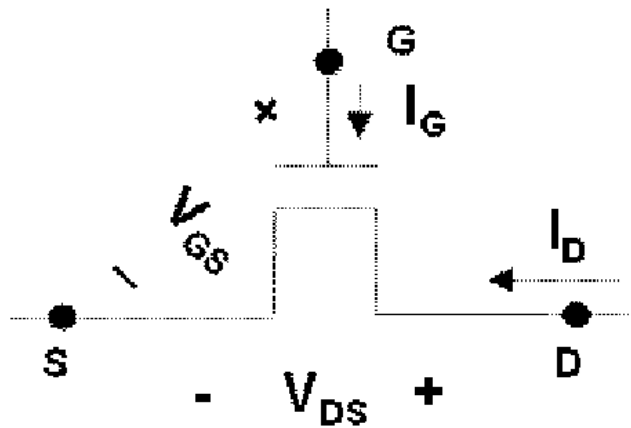
Setting V_{GS} above the threshold voltage is like connecting the water supply.

Cutoff = water supply disconnected (no path for current flow)

Setting V_{GS} to a larger value is like connecting a high-pressure water supply—more flow can potentially occur.

Water Tap Analogy

MOSFET Schematic

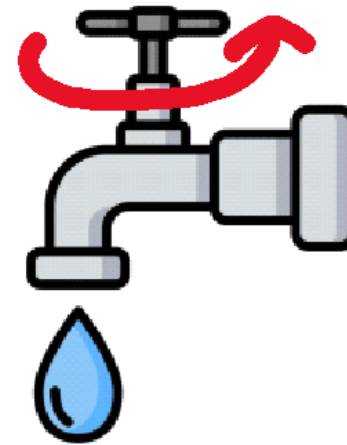


v_{GS} : Gate Voltage
 v_{DS} : Drain to Source Voltage

V_T : Threshold Voltage
 $V_{OV} = v_{GS} - V_T$: Overdrive voltage

Water Tap

Knob



Water Supply

Water Flow

Water Tap Analogy

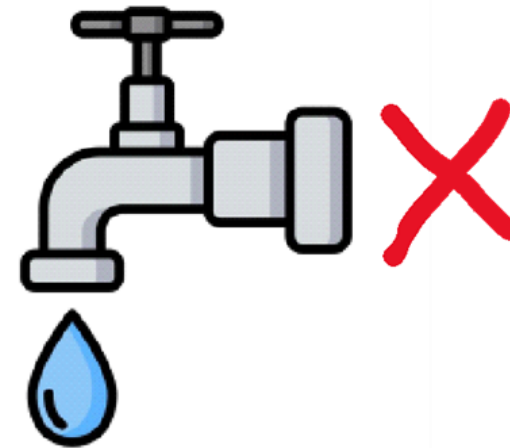
- The water tap itself is used to adjust water flow. You can turn the flow up and down.
- V_{DS} is like the water tap. It controls the amount of flow.
- There is, of course, a saturation point. If you keep turning the water tap control, eventually you won't get any more flow.
- **Triode = water tap in “normal range”, controls flow**
- **Saturation = water tap turned up to (or past) point for maximum flow**

Water Tap Analogy

Cutoff Mode

Occurs when

Water Supply Disconnected



Water Tap Analogy

Triode Mode

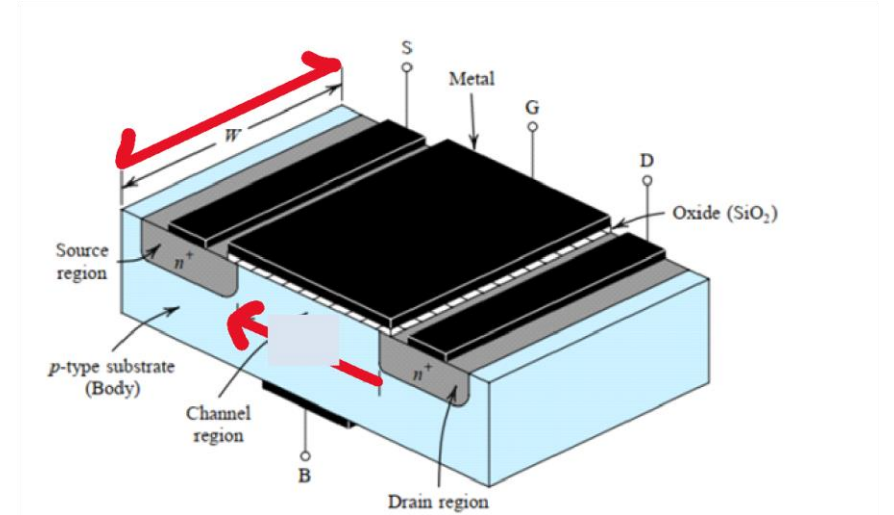
Occurs when _____ and _____

$$I_{DS} = k'_n \frac{W}{L} \left(v_{GS} - V_T - \frac{1}{2} v_{DS} \right) v_{DS}$$

$$k_n = k'_n \frac{W}{L}$$

MOSFET transconductance

Aspect Ratio



Water Supply **Connected**

- Water flow proportional to knob rotation



Water Tap Analogy

Saturation Mode

Occurs when _____ and _____

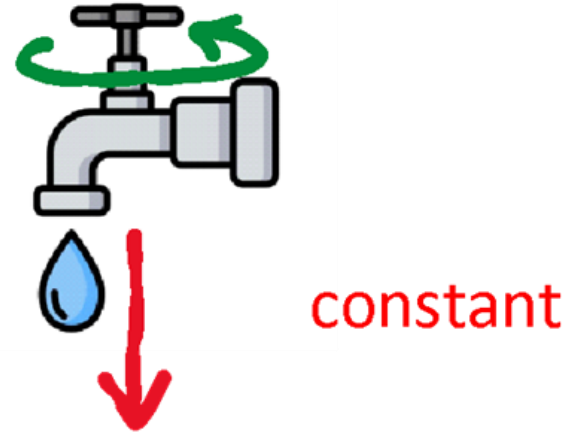
$$I_{DS} = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_T)^2$$

Overdrive Voltage is defined as:

$$V_{OV} = v_{GS} - V_T$$

Water Flow Saturated

- Water flow becomes constant and does not change with knob rotation



NMOS Equations

Cutoff Mode

Occurs when $v_{GS} < V_T$

$$I_D = 0$$

$v_{GS} - V_T = V_{OV}$: Overdrive Voltage

Triode Mode

Occurs when $v_{GS} \geq V_T$ and $v_{DS} < v_{GS} - V_T$

$$I_{DS} = k'_n \frac{W}{L} \left(v_{GS} - V_T - \frac{1}{2} v_{DS} \right) v_{DS}$$

V_{OV}
Indicates the available excess voltage at **gate** after forming the n-channel.

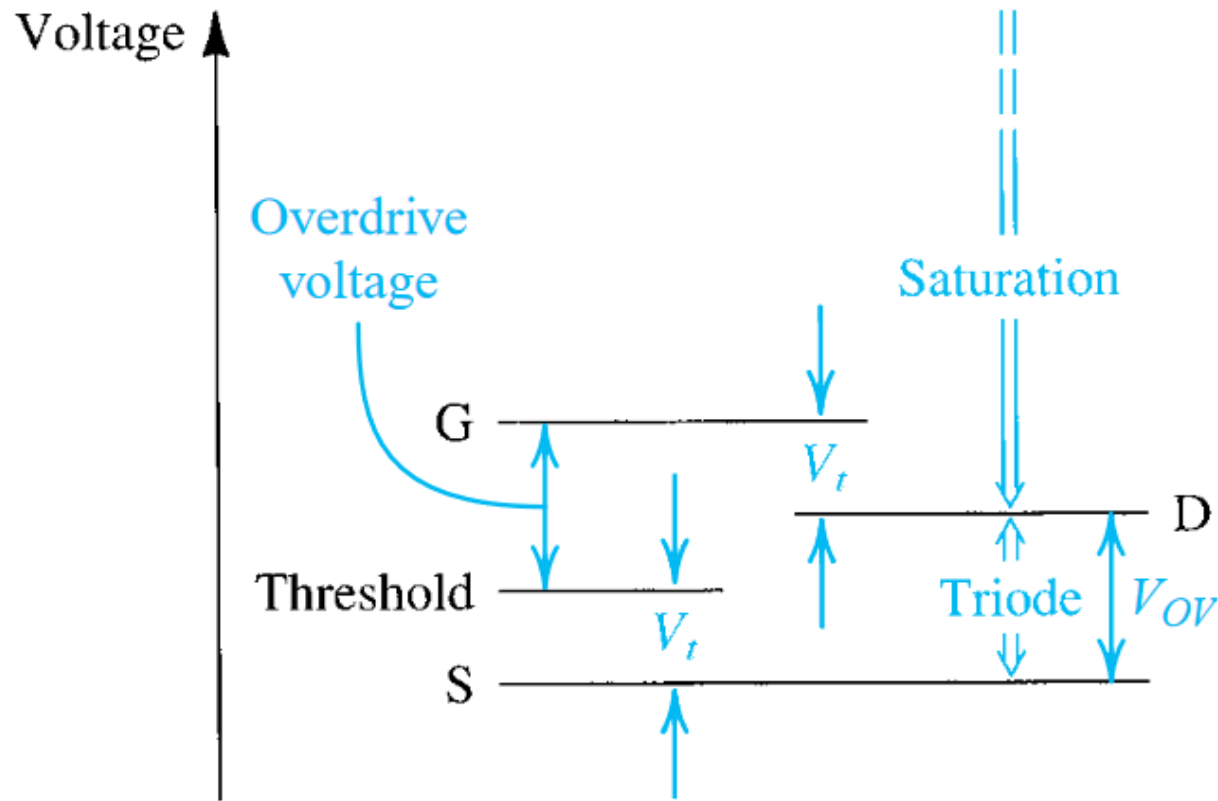
Saturation Mode

Occurs when $v_{GS} > V_T$ and $v_{DS} \geq v_{GS} - V_T$

$$I_{DS} = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_T)^2$$

V_T
The minimum voltage necessary at the **gate** terminal to form a channel.

Real MOSFET



$$k_n = k'_n \frac{W}{L}$$

MOSFET
transconductance

v_{GS} :

Gate Voltage

v_{DS} :

Drain to Source Voltage

V_T :

Threshold Voltage

$V_{OV} = v_{GS} - V_T$:

Overdrive voltage

Solving Circuits with MOSFET

- Use **Method of Assumed State!**
- Three steps:
 - **Assume:** One of the modes (Cutoff, Triode, Saturation)
 - **Solve:** Use corresponding equation and KCL+KVL
 - **Verify:** Check if the conditions of V_{DS} and V_{GS} are satisfied. If not, repeat.
- Might need to solve quadratic equation ($ax^2 + bx + c = 0$).
- If we get two roots, choose the one that's *favorable* to your assumption

Example 1

Analyze the circuit to find i_D and v_{o2} using the Method of Assumed State. Here, the input of the MOSFET is $v_{o1} = 1\text{ V}$. You must validate your assumptions.

Assume:

Let **MOSFET** be in **Saturation** mode:

$$v_{o1} > V_T$$

$$v_{DS} \geq v_{GS} - V_T \text{ or } v_{o2} \geq 0.8\text{ V}$$

Solve:

$$v_{o1} - V_T = 1 - 0.2 = 0.8\text{ V}$$

Current Equation: $I_D = \frac{k}{2}(v_{GS} - V_T)^2 = \frac{4}{2}(0.8)^2 = 1.28\text{ mA}$

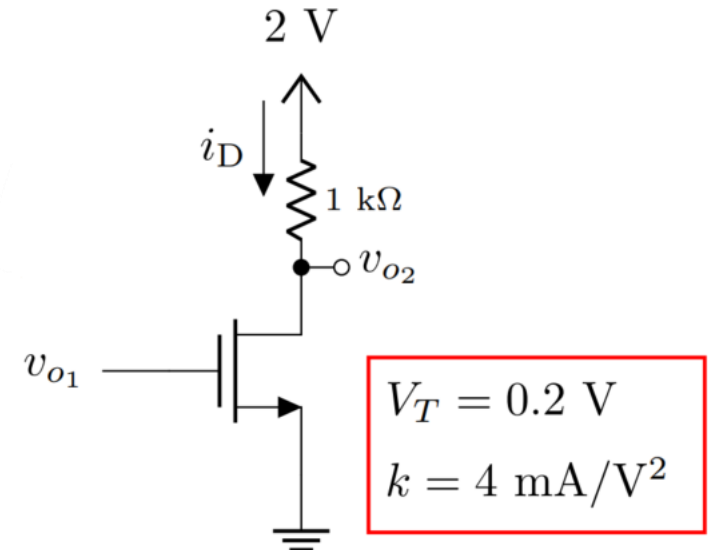
$$v_{o2} = 2 - I_D = 0.72\text{ V}$$

Verify: For **Saturation** mode $\rightarrow v_{o2} \geq 0.8\text{ V}$

But here, $v_{o2} = 0.72\text{ V} < 0.8\text{ V}$

Here, the conditions are **NOT** fulfilled.

Assumption is INCORRECT!



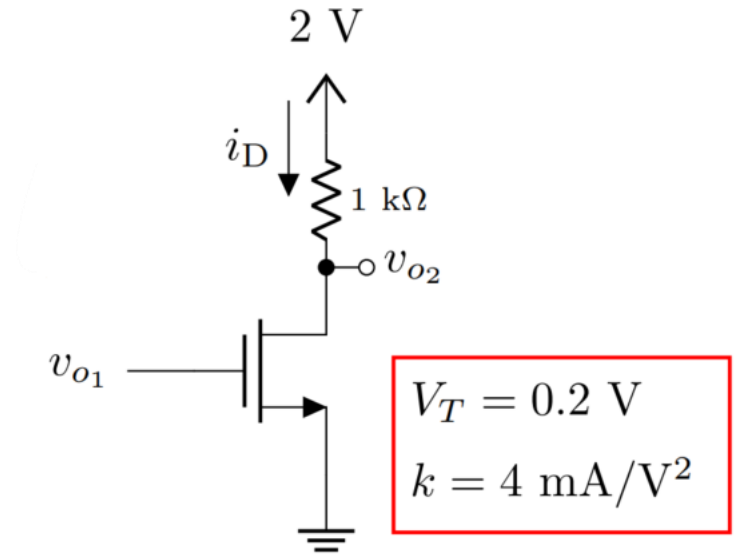
Example 1

Analyze the circuit to find i_D and v_{o2} using the Method of Assumed State. Here, the input of the MOSFET is $v_{o1} = 1\text{ V}$. You must validate your assumptions.

$\therefore$ MOSFET is in Triode Mode.

$$I_D = k \left(v_{GS} - V_T - \frac{1}{2} v_{DS} \right) v_{DS} = \frac{2 - v_{DS}}{1}$$

$$4 \left(0.8 - \frac{1}{2} v_{o2} \right) v_{o2} = 2 - v_{o2}$$
$$v_{o2} = \cancel{1.37\text{ V}} \text{ or } 0.73\text{ V}$$



Example 2

- Design the circuit, that is, determine the values of R_D and R , so that the transistor operates at $I_D = 0.4 \text{ mA}$ and $V_D = +0.5 \text{ V}$. The NMOS transistor has $V_T = 0.7 \text{ V}$, $k = 3.2 \text{ mA/V}^2$.

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{2.5 - 0.5}{0.4} = 5 \text{ k}\Omega$$

Assume:

Let **MOSFET** be in **Saturation** mode:

$$v_{GS} > V_T$$

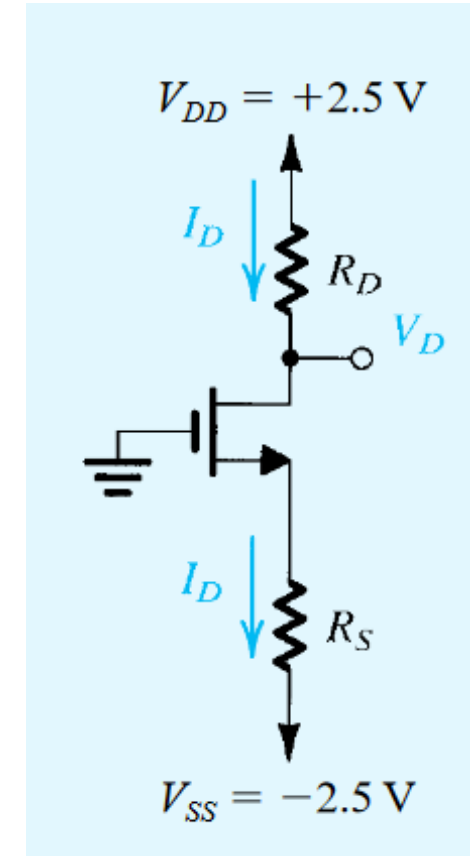
$$v_{DS} \geq v_{GS} - V_T$$

Solve:

$$\begin{aligned} v_{GS} &= -V_S \\ V_{GS} - V_T &= -V_S - 0.7 \end{aligned}$$

$$\text{Current Equation: } I_D = \frac{k}{2} (v_{GS} - V_T)^2 = \frac{3.2}{2} (-V_S - 0.7)^2 = 0.4 \text{ mA}$$

$$V_S = -0.2 \text{ V or } -1.2 \text{ V}$$



Example 2

- Design the circuit, that is, determine the values of R_D and R , so that the transistor operates at $I_D = 0.4 \text{ mA}$ and $V_D = +0.5 \text{ V}$. The NMOS transistor has $V_T = 0.7 \text{ V}$, $k = 3.2 \text{ mA/V}^2$.

Verify:

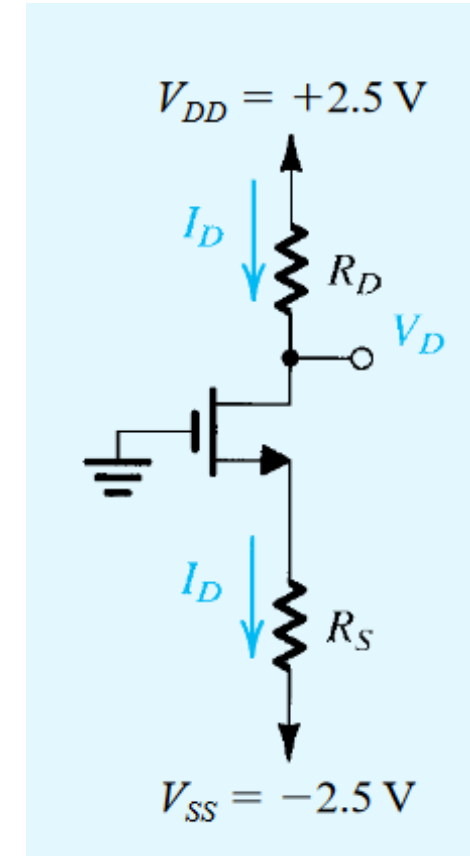
$$V_{DS} = 0.5 - V_S = 1.7 \text{ V}$$
$$V_{GS} - V_T = -V_S - 0.7 = 0.5 \text{ V}$$

Clearly

$$V_{DS} \geq V_{GS} - V_T$$

Assumption is CORRECT!

$$R_S = \frac{V_S - V_{SS}}{I_D} = \frac{-1.2 - (-2.5)}{0.4} = 5.75 \text{ k}\Omega$$



Example - 3

For the circuit, find the value of R that results in $V_D = 0.8$ V. The MOSFET has

$$V_T = 0.5 \text{ V}, k'_n = \mu_n C_{OX} = 0.4 \text{ mA/V}^2, \frac{W}{L} = \frac{0.72 \mu\text{m}}{0.18 \mu\text{m}}$$

$$k_n = k'_n \frac{W}{L} = 1.6 \text{ mA/V}^2$$

Assume:

Let MOSFET be **IS** in **Saturation** mode:

Proof:

$$V_D = V_G$$

$$V_D - V_S = V_G - V_S$$

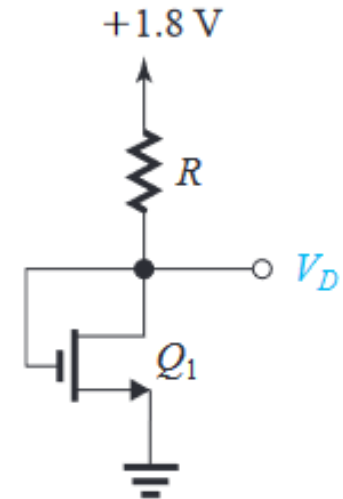
$$V_{DS} = V_{GS} \geq V_{GS} - V_T \rightarrow \text{Saturation Mode}$$

Solve:

$$V_{GS} - V_T = V_{DS} - V_T = 0.8 - 0.5 = 0.3 \text{ V}$$

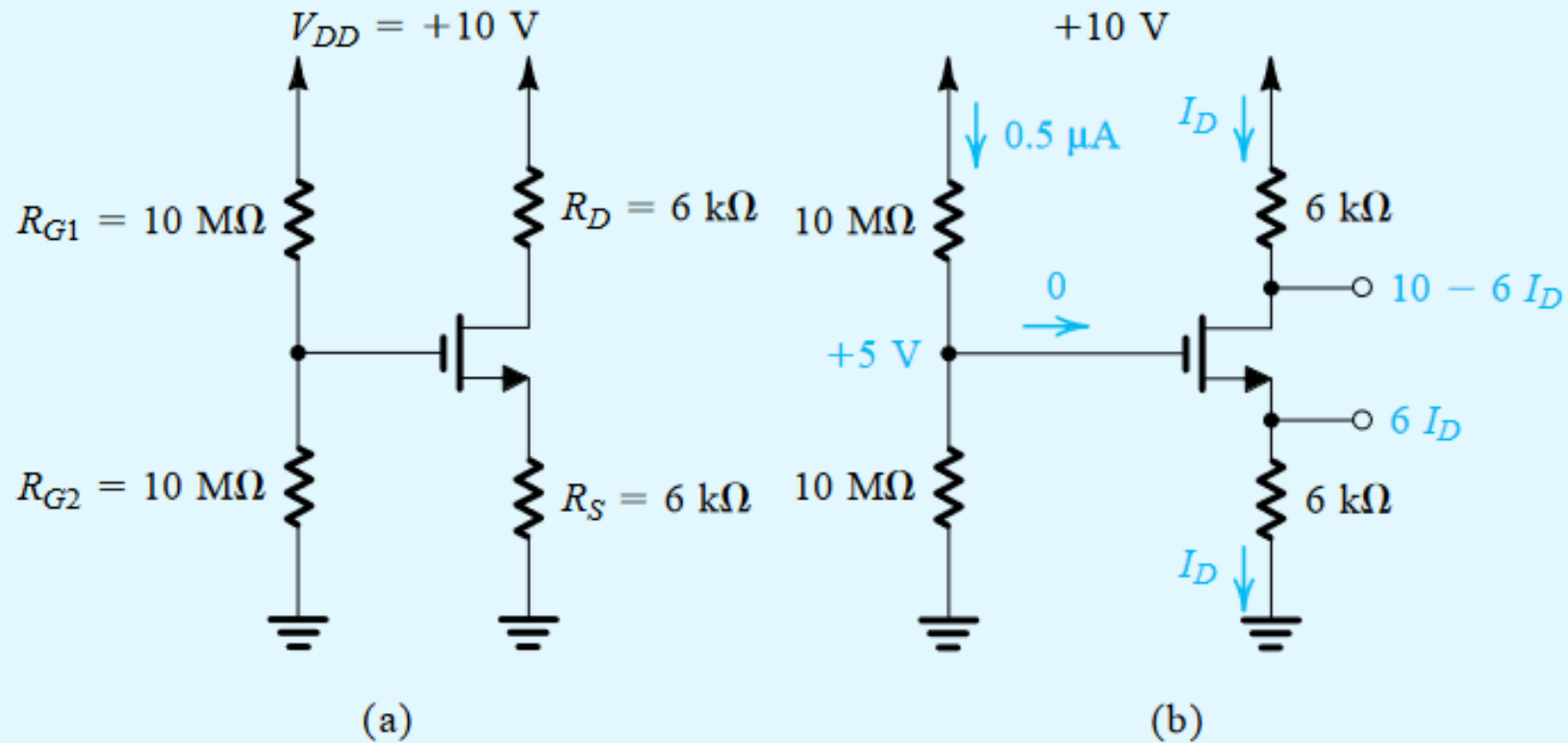
$$\text{Current Equation: } I_D = \frac{k_n}{2} (V_{GS} - V_T)^2 = \frac{1.6}{2} (0.3)^2 = 0.072 \text{ mA}$$

$$R_D = \frac{1.8 - V_D}{I_D} = \frac{1.8 - 0.8}{0.072} = 13.89 \text{ k}\Omega$$



Example - 4

Analyze the circuit shown in Fig. 5.24(a) to determine the voltages at all nodes and the currents through all branches. Let $V_{tn} = 1$ V and $k'_n(W/L) = 1$ mA/V². Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$).



Example - 4

Analyze the circuit shown in Fig. 5.24(a) to determine the voltages at all nodes and the currents through all branches. Let $V_{tn} = 1$ V and $k'_n(W/L) = 1$ mA/V². Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$).

Assume:

Let **MOSFET** be in **Saturation** mode:

$$V_{GS} < V_T$$

$$V_{DS} \geq V_{GS} - V_T$$

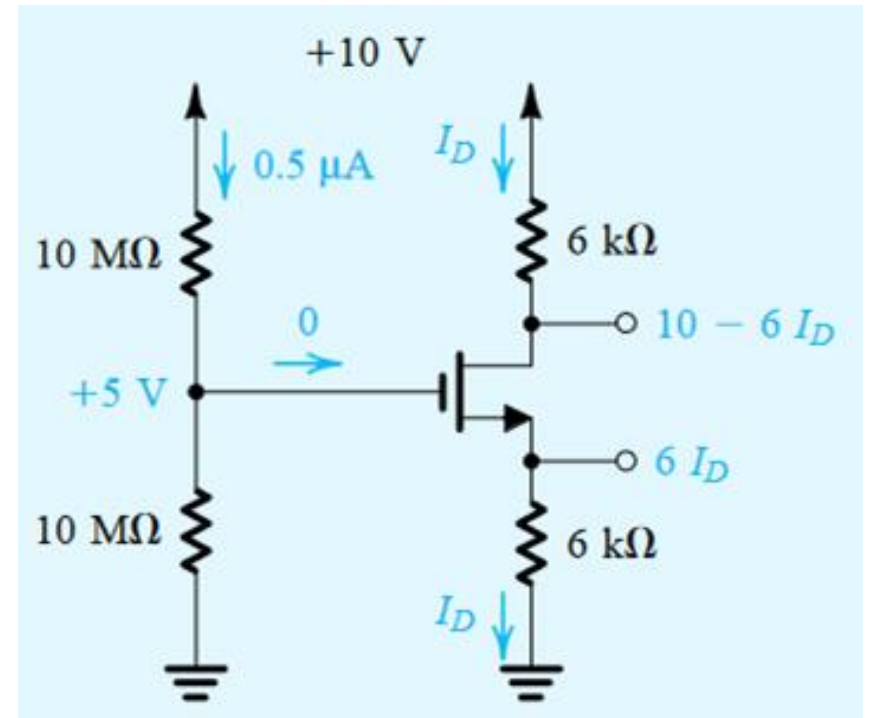
Solve:

$$I_D = \frac{10 - V_D}{6} = \frac{V_S}{6}$$

Current Equation: $I_D = \frac{k}{2}(V_{GS} - V_T)^2 = \frac{1}{2}(5 - V_S - 1)^2$

$$I_D = \frac{V_S}{6} = \frac{1}{2}(4 - V_S)^2$$

$$V_S = \cancel{5.33 \text{ V}} \text{ or } 3 \text{ V}$$



Example - 4

Analyze the circuit shown in Fig. 5.24(a) to determine the voltages at all nodes and the currents through all branches. Let $V_{tn} = 1$ V and $k'_n(W/L) = 1$ mA/V². Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$).

Verify:

$$I_D = \frac{V_S}{6} = \frac{3}{6} = 0.5 \text{ mA}$$

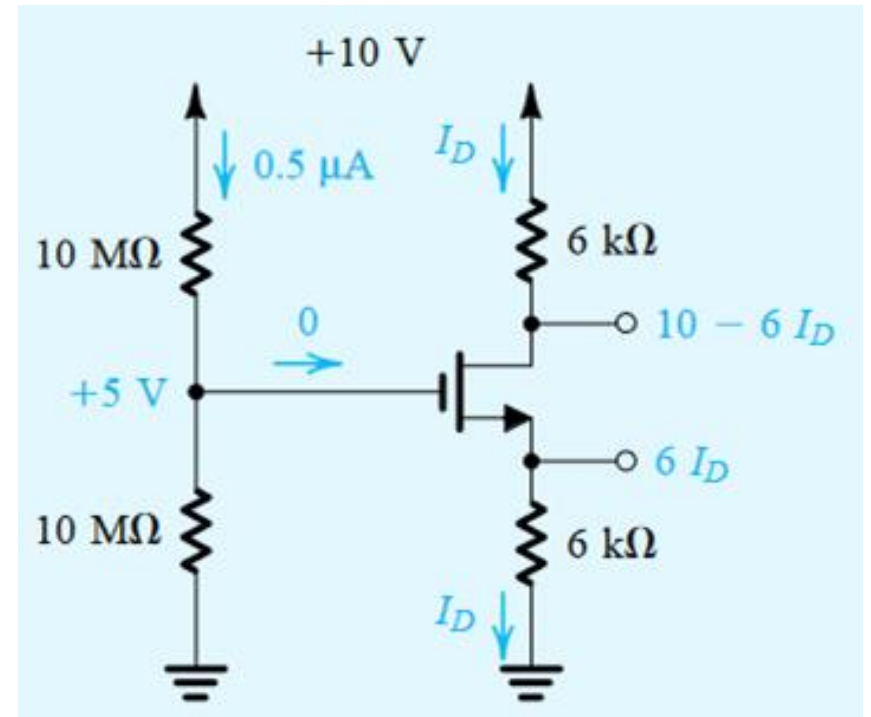
$$V_D = 10 - 6I_D = 7 \text{ V}$$

$$V_{DS} = V_D - V_S = 7 - 3 = 4 \text{ V}$$

$$V_{GS} - V_T = 4 - V_S = 1 \text{ V}$$

For **Saturation** mode $\rightarrow V_{DS} \geq V_{GS} - V_T$

Assumption is CORRECT!



Example - 5

D 5.50 The NMOS transistors in the circuit of Fig. P5.50 have $V_t = 0.5$ V, $\mu_n C_{ox} = 250$ $\mu\text{A}/\text{V}^2$, $\lambda = 0$, and $L_1 = L_2 = 0.25$ μm . Find the required values of gate width for each of Q_1 and Q_2 , and the value of R , to obtain the voltage and current values indicated.

Solve:

Both MOSFET are in SATURATION:

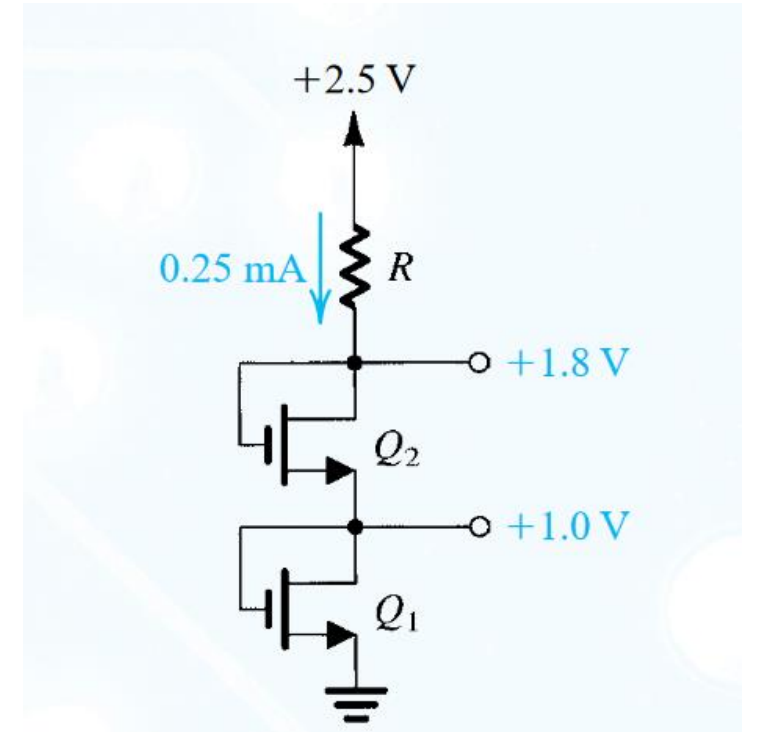
Q1: $V_{GS1} - V_T = 1 - V_T = 1 - 0.5 = 0.5$ V

Q2: $V_{GS2} - V_T = (1.8 - 1) - V_T = 0.8 - 0.5 = 0.3$ V

Current Equation:

$$I_D = \frac{\mu_n C_{ox}}{2} \left(\frac{W_1}{L_1} \right) (V_{GS1} - V_T)^2 = \frac{\mu_n C_{ox}}{2} \left(\frac{W_2}{L_2} \right) (V_{GS2} - V_T)^2 = 0.25 \text{ mA}$$

$$\frac{0.25}{2} \left(\frac{W_1}{0.25} \right) (0.5)^2 = \frac{0.25}{2} \left(\frac{W_2}{0.25} \right) (0.3)^2 = 0.25$$



Example - 5

D 5.50 The NMOS transistors in the circuit of Fig. P5.50 have $V_t = 0.5$ V, $\mu_n C_{ox} = 250$ $\mu\text{A}/\text{V}^2$, $\lambda = 0$, and $L_1 = L_2 = 0.25$ μm . Find the required values of gate width for each of Q_1 and Q_2 , and the value of R , to obtain the voltage and current values indicated.

$$R = \frac{2.5 - 1.8}{I_D} = 2.8 \text{ k}\Omega$$

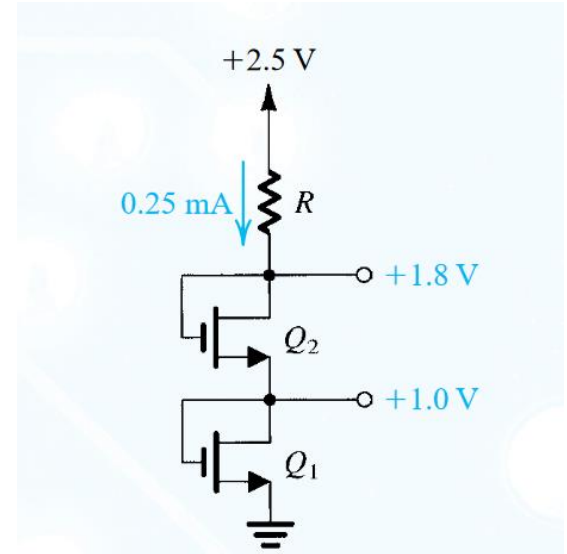
Solve:

$$I_D = \frac{\mu_n C_{ox}}{2} \left(\frac{W_1}{L_1} \right) (V_{GS1} - V_T)^2 = \frac{\mu_n C_{ox}}{2} \left(\frac{W_2}{L_2} \right) (V_{GS2} - V_T)^2 = 0.25 \text{ mA}$$
$$\frac{0.25}{2} \left(\frac{W_1}{0.25} \right) (0.5)^2 = \frac{0.25}{2} \left(\frac{W_2}{0.25} \right) (0.3)^2 = 0.25$$

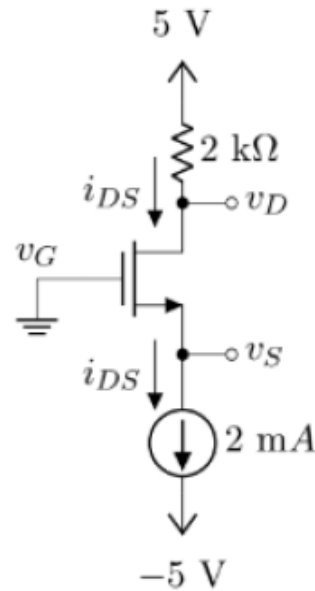
$$W_1 \times 0.25 = W_2 \times 0.09 = 2 \times 0.25$$

$$\frac{W_1}{4} = \frac{9W_2}{100} = \frac{1}{2}$$

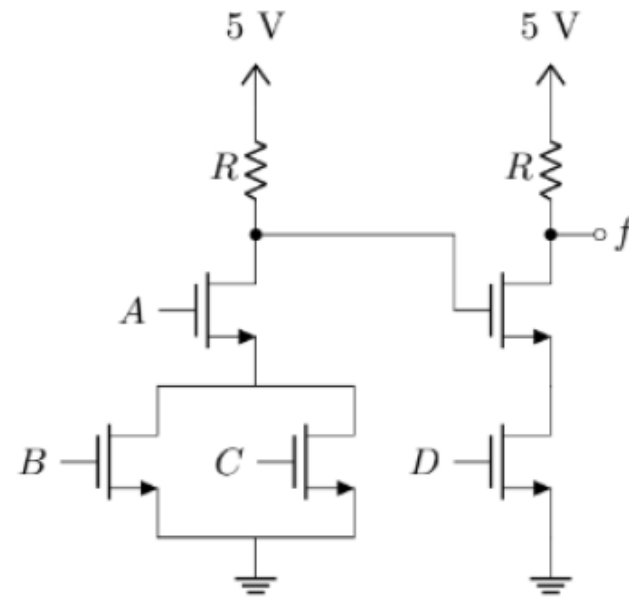
$$W_1 = 2 \text{ } \mu\text{m} \text{ or } W_2 = 5.556 \text{ } \mu\text{m}$$



Problem 1



Circuit 1



Circuit 2

Refer to the circuit-1 above. For the MOSFET, $V_T = 1\text{ V}$ and $K = 4\text{ mA/V}^2$.

a) **Identify** the value of the gate voltage V_G and the drain-source current i_{DS} .

b) **Calculate** the value of the drain voltage V_D using the $2\text{ k}\Omega$ resistor.

c) **Analyze** the circuit to find V_S . Here, use the Method of Assumed State. You must validate your assumptions.

d) **Analyze** the Circuit 2 above to find f in terms of boolean inputs A, B, C, and D.

$$V_D = 1\text{ V}$$

$$V_{GS} = 2\text{ V}$$

$$V_S = -2\text{ V}$$

$$V_{ov} = 1\text{ V}$$

Problem 2

In the circuit above, the MOSFETs have the following parameters, $k_n' = 2 \text{ mA/V}^2$, $W/L = 2.5$, $V_T = 0.5 \text{ V}$.

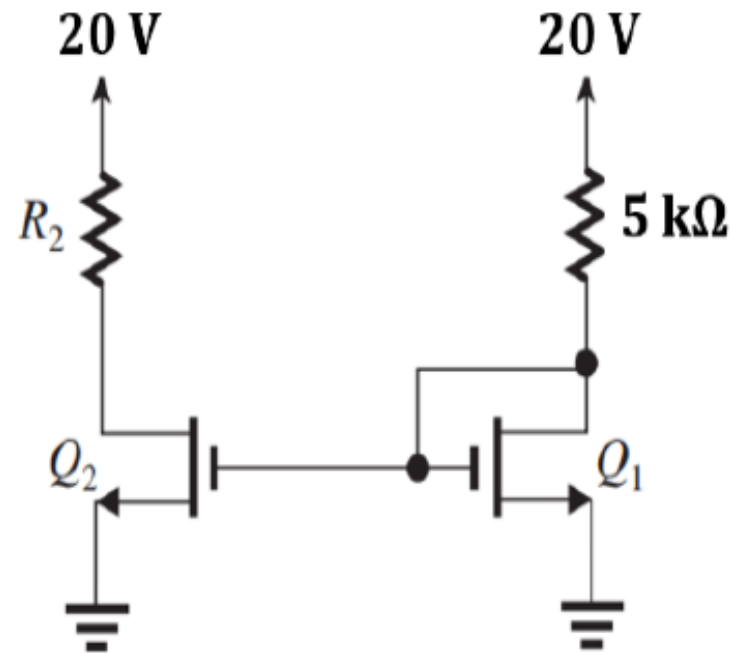
- a) **Find out** the operating mode of Q_1
- b) **Determine** the value of R_2 that results in Q_2 operating at the edge of the saturation region.
- c) **Calculate** the on-state resistance, R_{on} for Q_2

$$V_{G1} = 1.71 \text{ V or } -0.79 \text{ V}$$

$$V_{G1} = V_{G2} = 1.71 \text{ V}$$

$$V_{DS2} = 1.21 \text{ V}, V_D = 1.21 \text{ V} \quad I_{DS2} = 3.66 \text{ mA}$$

$$R_2 = 5.134 \text{ k}$$



Problem 3

Find I_o and V_o

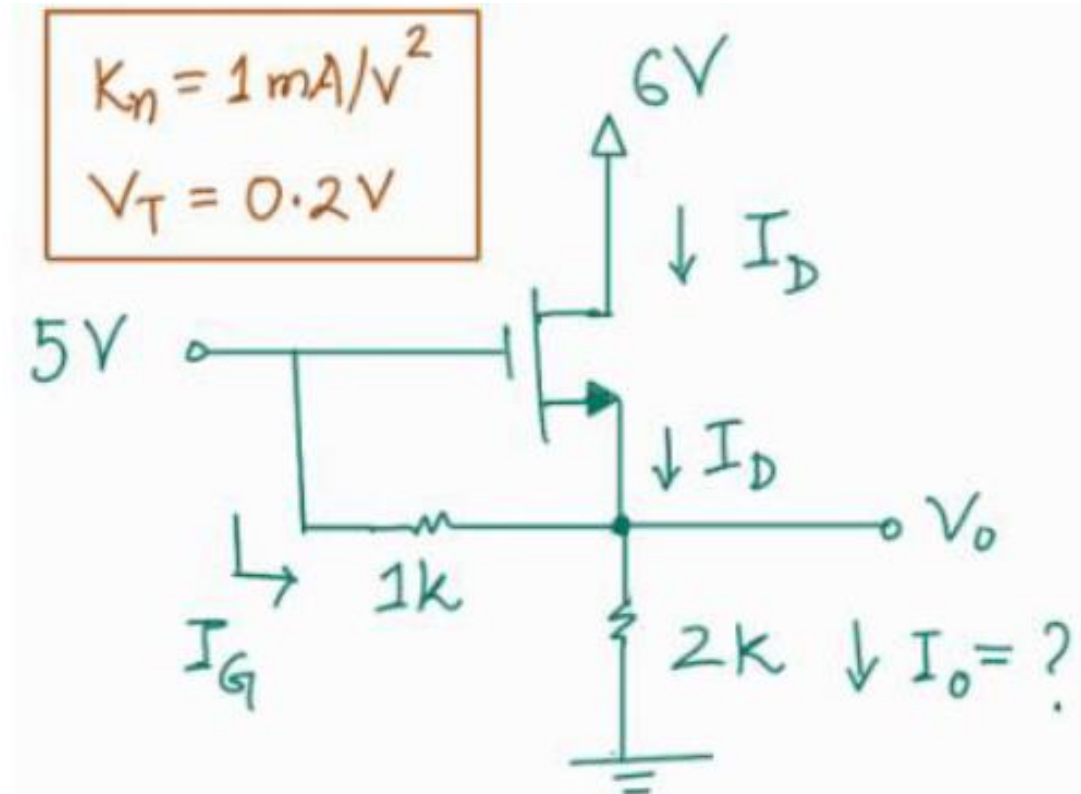
$$V_s = 3.72\text{V or } \cancel{8.87\text{V}}$$

$$V_{GS} = 1.26\text{V}$$

$$V_{ov} = 1.06\text{V}$$

$$V_{DS} = 2.28\text{V}$$

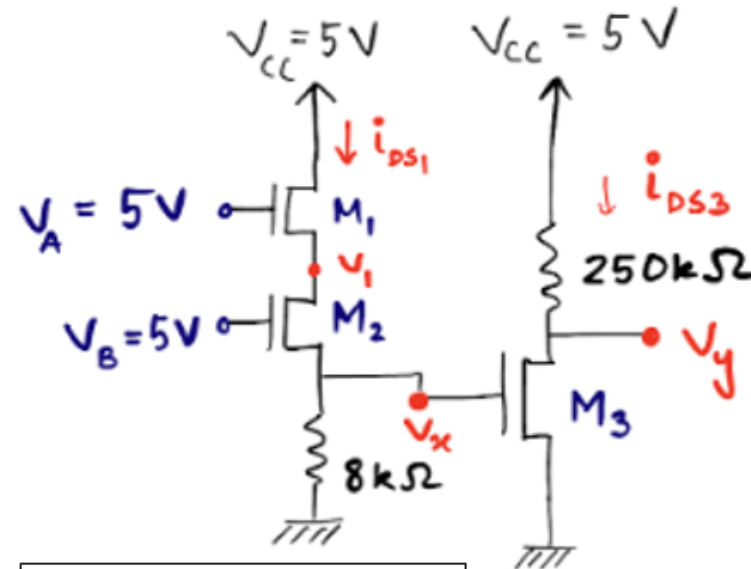
$$I_o = 1.86\text{mA}$$



Problem 4

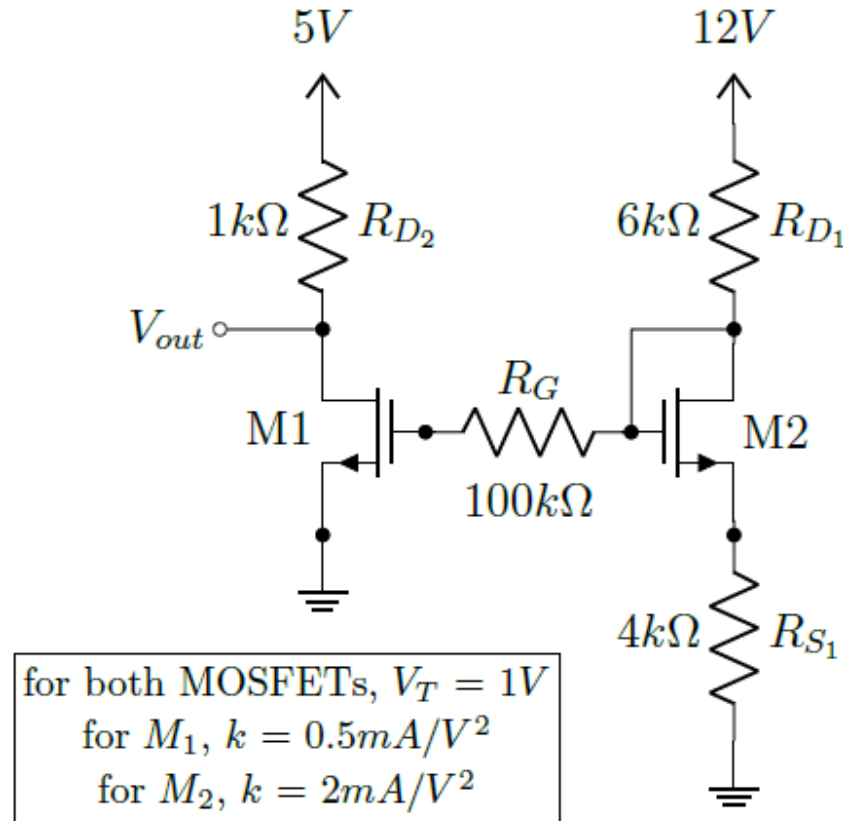
For the following circuit with MOSFETs, assume that $V_T = 1V$, $k_n' = 50 \mu A/V^2$ and the aspect ratio W/L is 1 for all the MOSFETS.

- a) If the gate voltage to M_3 MOSFET, $V_x = 0.938V$, and MOSFET M_2 is in triode mode, **find** the voltage V_1 .
- b) **Find** V_y .
- c) If $W/L=2$, then $V_1 = 2.1454 V$. **Find** V_x for this case.



$$\begin{aligned} V_1 &= 1.835V \text{ or } \cancel{6.16V} \\ V_{DS1} &= 3.165V \\ V_{GS1} &= 3.165V \\ V_{ov} &= 2.165V \\ V_y &= 5V \\ V_x &= 1.377V \end{aligned}$$

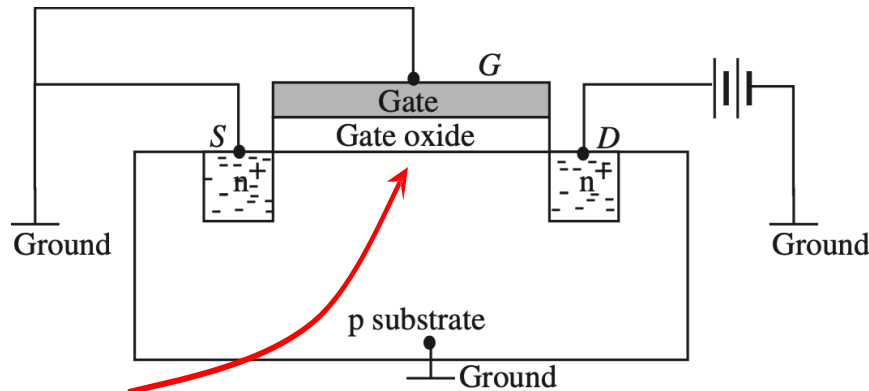
Problem 5



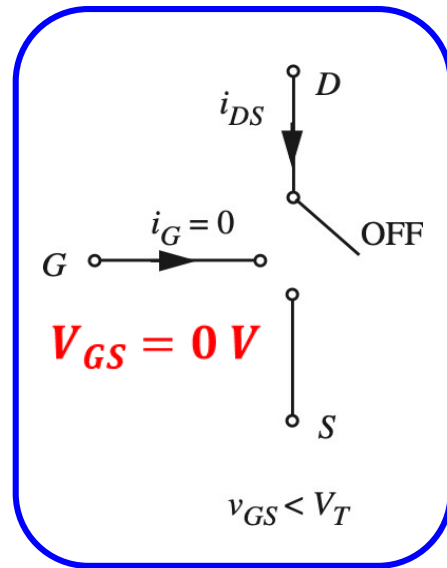
$$\begin{aligned} V_{S2} &= 4V, \quad \cancel{4.84V} \\ V_{D2} &= 6V \\ V_{G2} &= 6V \\ I_{RD1} &= 1mA \\ V_{G1} &= 6V \\ V_{D1} &= 1.615V, \quad \cancel{12.385V} \\ I_{RD2} &= 3.385mA \end{aligned}$$

- (b) [6 marks] Analyze the circuit in *Figure-2* and answer the following questions:
- Show that, if the MOSFET, M2 conducts current, it will operate in the saturation mode. [Hint: You don't need to solve the circuit]
 - Calculate the values of V_{out} , I_{RD1} , and I_{RD2} using the method of assuming states. You must validate your assumptions.

n-channel MOSFET - Summary

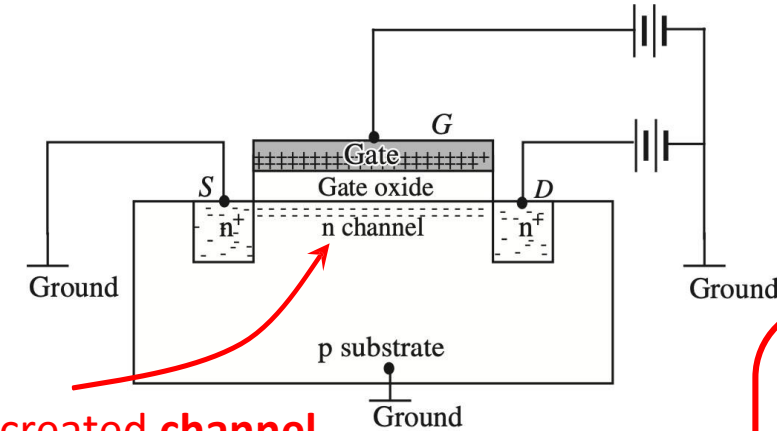


No channel, open ckt



Cut off mode

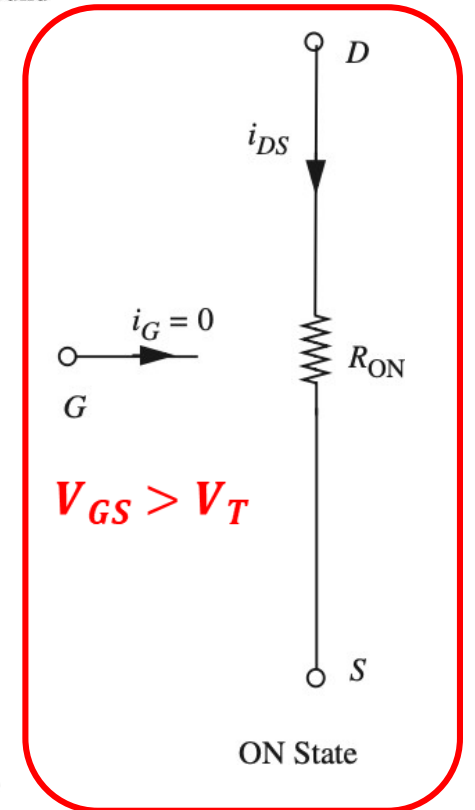
G – Gate:
S – Source:
D – Drain:



The created **channel**
will have some R
-> SR model

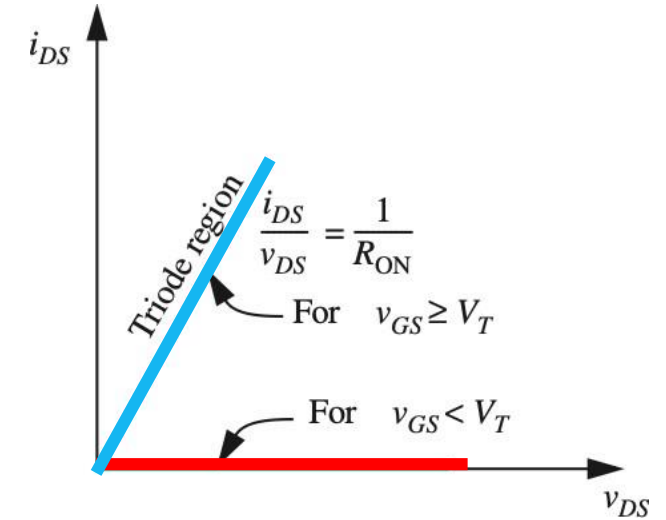
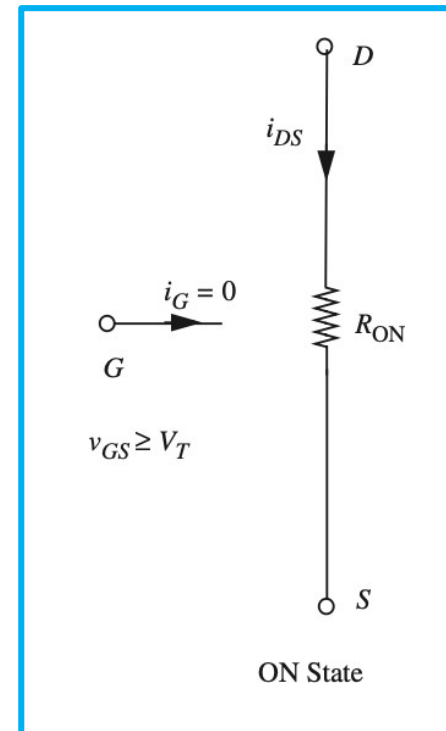
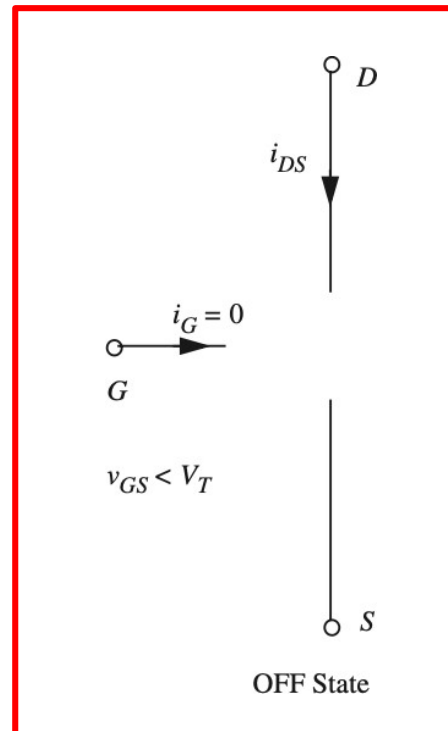
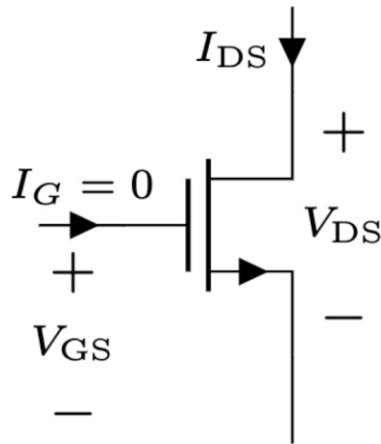
Trigger terminal
Charge carrier reservoir
Charge carrier sink

Small v_{DS} : Triode Mode
Large v_{DS} : Saturation Mode



n-channel MOSFET - Summary

SR Model:

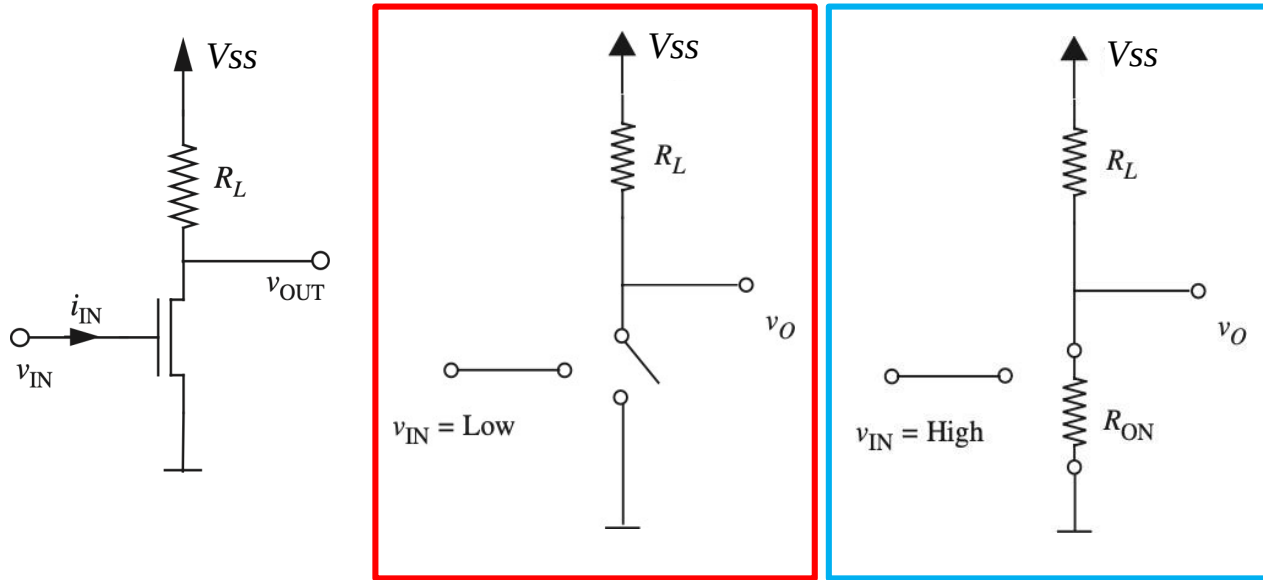


$$R_{ON} = \frac{1}{k_n(V_{OV})}$$

V_{OV}
Indicates the available
excess voltage at **gate** after
forming the n-channel.

- SR model is a better approximation than S model.
- **Triode Mode exists until:** $v_{DS} \leq (v_{GS} - V_T)$

SR Model - Inverter



$$v_{o,HI} = V_{SS}$$

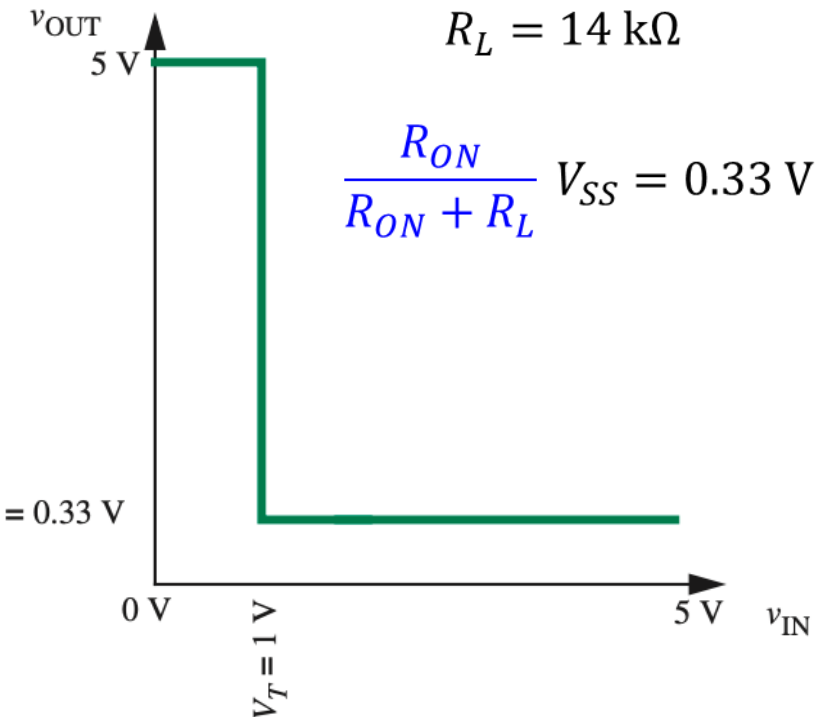
$$v_{o,LO} = \frac{R_{ON}}{R_{ON} + R_L} V_{SS}$$

For example, If

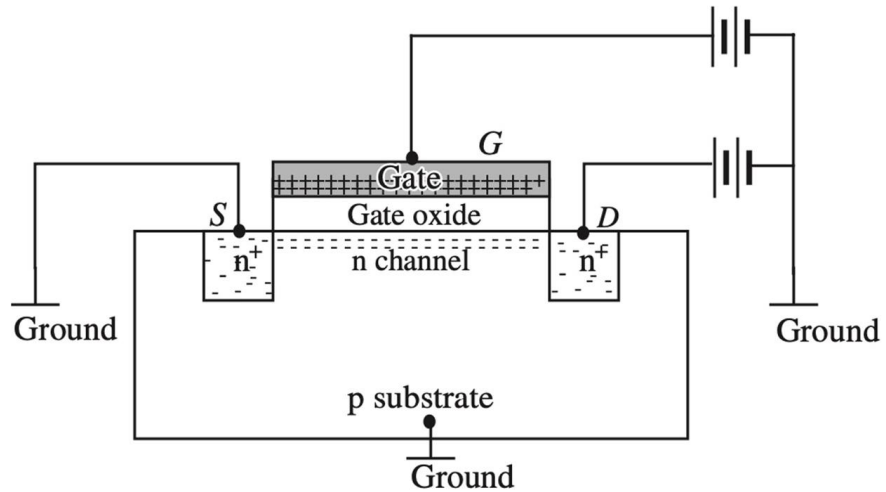
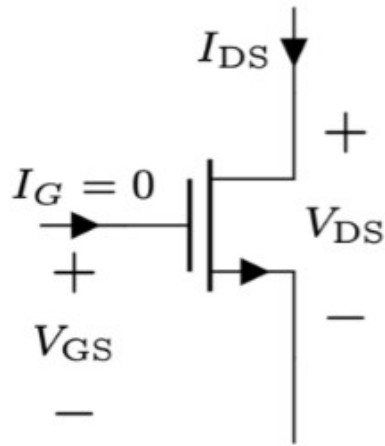
$$\begin{aligned} V_{SS} &= 5 \text{ V}, \\ R_{ON} &= 1 \text{ k}\Omega, \\ R_L &= 14 \text{ k}\Omega \end{aligned}$$

$$\frac{R_{ON}}{R_{ON} + R_L} V_{SS} = 0.33 \text{ V}$$

$$\frac{V_{SS} R_{ON}}{R_{ON} + R_L} = 0.33 \text{ V}$$



Review – MOSFET



Control = $V_{GS} = V_G - V_S$ controls the IV between drain-source (I_{DS} vs V_{DS})

Threshold voltage = V_T , minimum voltage required to create the channel

Models

1. **S Mode:** Assumes an ideal channel with zero resistance
2. **SR Model:** Assumes finite channel resistance, R_{ON} , depends on $V_{GS} - V_T = V_{OV}$

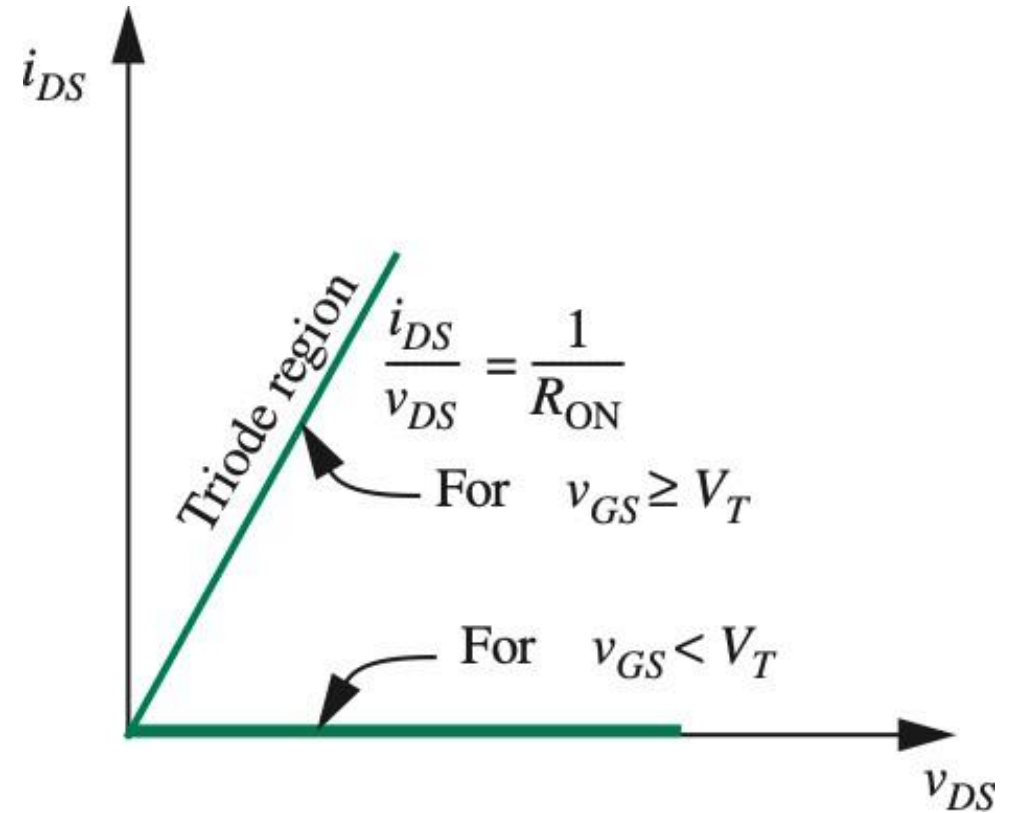
NMOS Triode Mode

For small v_{DS}

$$I_{DS} = k'_n \frac{W}{L} \left(v_{GS} - V_T - \frac{1}{2} v_{DS} \right) v_{DS}$$

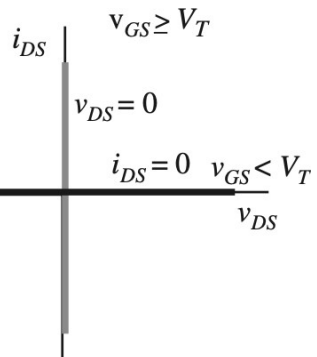
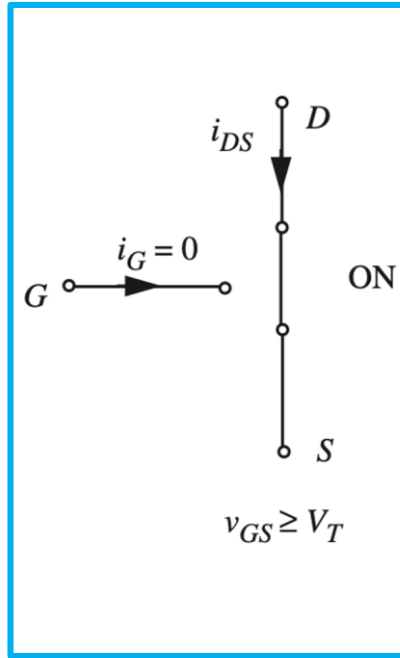
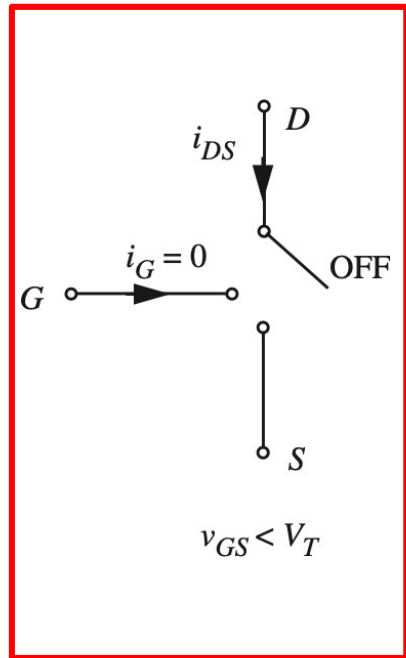
$$I_{DS} = \frac{1}{R_{ON}} v_{DS}$$

$$R_{ON} = \frac{1}{k'_n \frac{W}{L} (v_{GS} - V_T)}$$
$$= \frac{1}{k_n (V_{OV})}$$

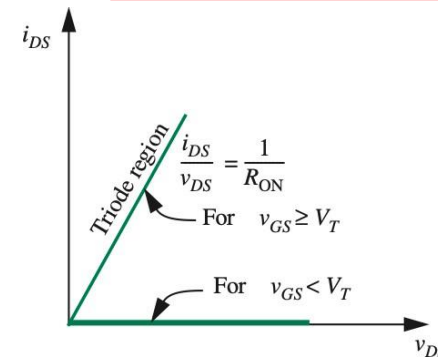
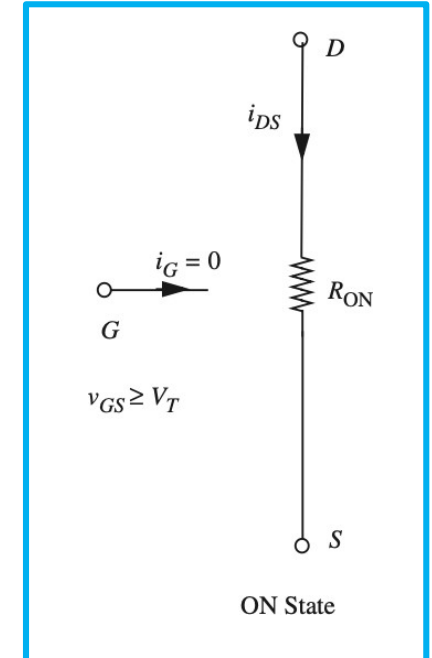
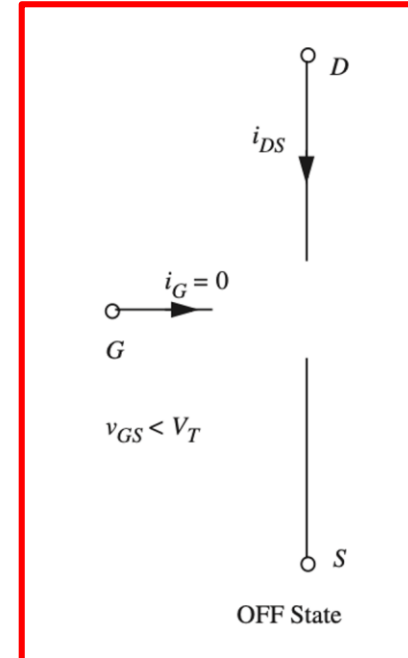


MOSFET Linear Models

S Model



SR Model



$$R_{ON} = \frac{1}{k'_n \frac{W}{L} (v_{GS} - V_T)}$$